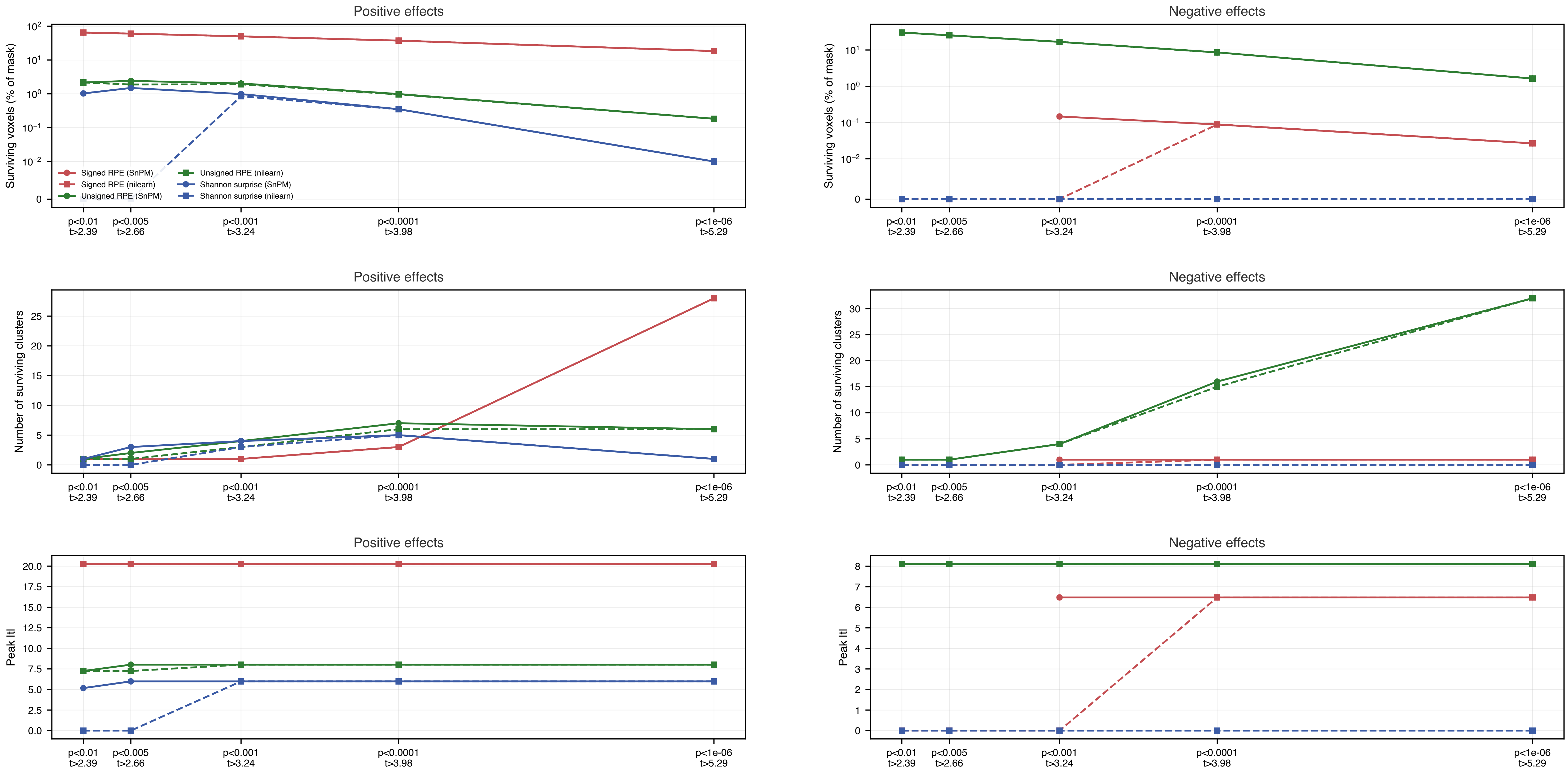
How much of the brain survives — model7

Extent of the surviving map as a function of the cluster-forming threshold. If a signal's advantage is sensitivity rather than anatomy, it shows up here as a curve that sits above the others at every threshold.



Engine comparison and caveats

SnPM and nilearn run the same sign-flipping permutation test on the same 58 contrast images and the same analysis mask (SnPM's own), but they are not identical, and three differences matter for reading the pages that follow:

1. Family. SnPM's `Tsign` treats each tail as its own one-tailed family, which is what the paper reports; nilearn's `two_sided_test=True` controls the family-wise error over both tails at once. nilearn is therefore the stricter test by roughly a factor of two in the tail, and its maps should be slightly smaller wherever a threshold is near the edge of significance.
2. Connectivity. SPM labels clusters with 18-connectivity (`spm_clusters`); nilearn hardcodes 6-connectivity. These runs monkeypatch nilearn to 18 so the two engines parcellate identically -- otherwise nilearn splits a single SPM cluster into several smaller ones, each of which is then easier to kill on extent.
3. Statistic. SnPM's cluster-extent test uses cluster size only. nilearn additionally provides cluster *mass* (the summed excess of t over the threshold) and TFCE, both of which reward tall-and-narrow clusters that pure extent penalises.

TFCE and voxel-wise FWE are threshold-free and appear once, not once per cluster-forming threshold. Where the two engines disagree, the Dice overlap on the summary panels is the place to look first; large disagreement almost always means the map is sitting on the edge of the extent criterion, which is itself the point of this document.

Across all 23 signal × sign × threshold panels in this document, the SnPM and nilearn survivor maps overlap with a median Dice of 1.00 (range 0.00–1.00).

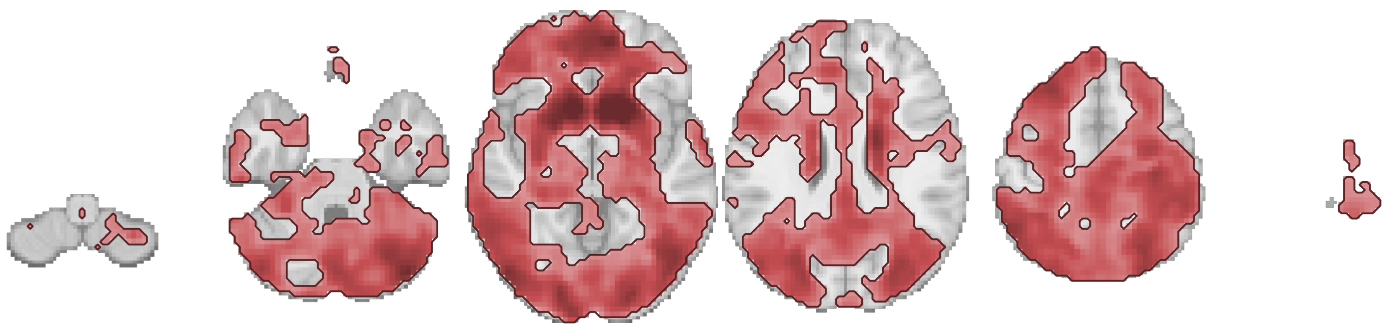
SnPM cluster-extent FWE — model7

Cluster-forming t > 2.3936 (one-tailed p < 0.01, df = 57); clusters with p_FWE < .05, 5000 permutations, one-tailed family per tail. Each slice is cut through one cluster's peak.

▲ Signed RPE — Positive

One component covers the map — cluster-extent inference is uninformative here

1 connected component (18-conn) · 11 local maxima ≥8 mm apart · 38,852 voxels (64.9% of mask) · 1,223.8 cm³ · peak |t| = 20.3



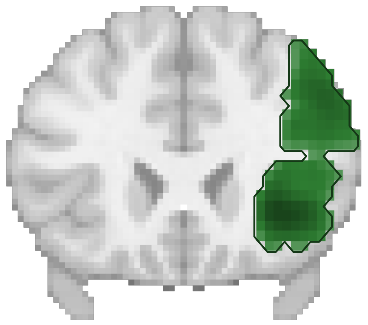
#1 1,223,838 mm³ t=+20.3
6 axial cuts peak (14, 14, -8)

▼ Signed RPE — Negative

Map not available

▲ Unsigned RPE — Positive

1 connected component (18-conn) · 4 local maxima ≥8 mm apart · 1,299 voxels (2.2% of mask) · 40.9 cm³ · peak |t| = 7.2

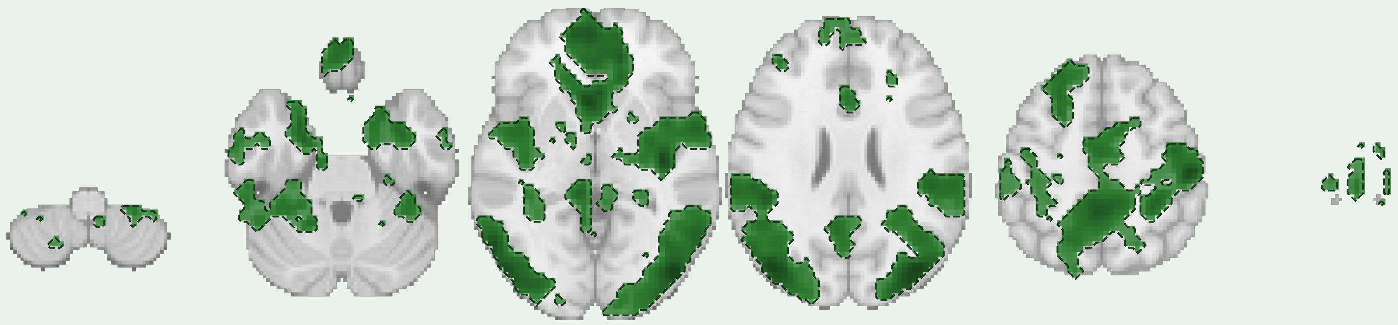


#1 40,918 mm³ t=+7.2
y=24 peak (36, 24, -5)

▼ Unsigned RPE — Negative

One component covers the map — cluster-extent inference is uninformative here

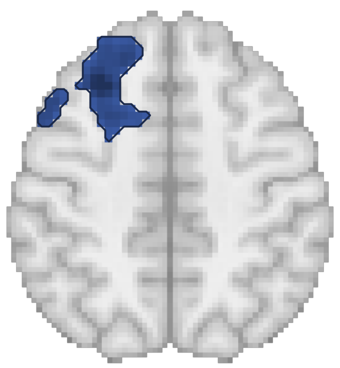
1 connected component (18-conn) · 17 local maxima ≥8 mm apart · 18,112 voxels (30.3% of mask) · 570.5 cm³ · peak |t| = 8.1



#1 570,528 mm³ t=-8.1
6 axial cuts peak (2, 18, -8)

▲ Shannon surprise — Positive

1 connected component (18-conn) · 13 local maxima ≥8 mm apart · 616 voxels (1.0% of mask) · 19.4 cm³ · peak |t| = 5.2



#1 19,404 mm³ t=+5.2
z=48 peak (-28, 24, 48)

▼ Shannon surprise — Negative

Map not available

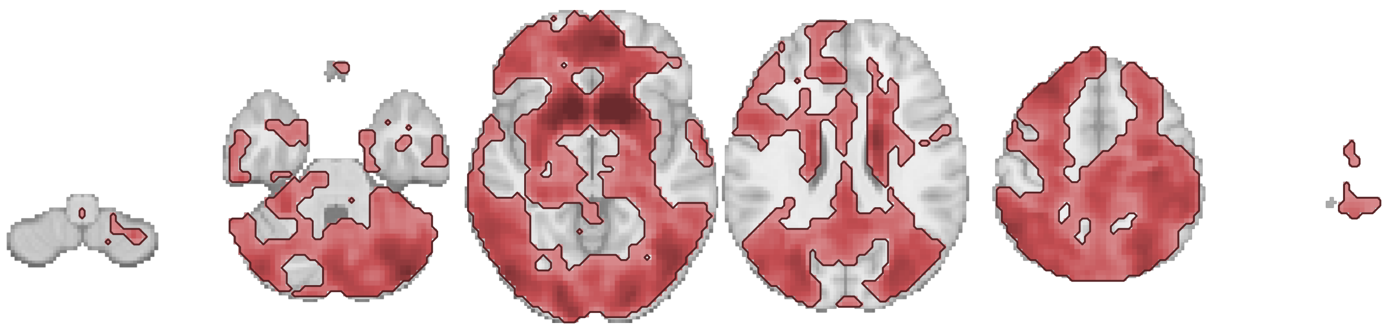
SnPM cluster-extent FWE — model7

Cluster-forming t > 2.6649 (one-tailed p < 0.005, df = 57); clusters with p_FWE < .05, 5000 permutations, one-tailed family per tail. Each slice is cut through one cluster's peak.

▲ Signed RPE — Positive

One component covers the map — cluster-extent inference is uninformative here

1 connected component (18-conn) · 8 local maxima ≥8 mm apart · 36,097 voxels (60.3% of mask) · 1,137.1 cm³ · peak |t| = 20.3



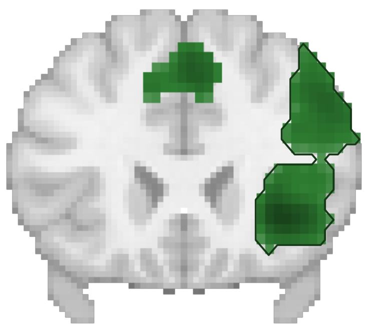
#1 1,137,056 mm³ t=+20.3
6 axial cuts peak (14, 14, -8)

▼ Signed RPE — Negative

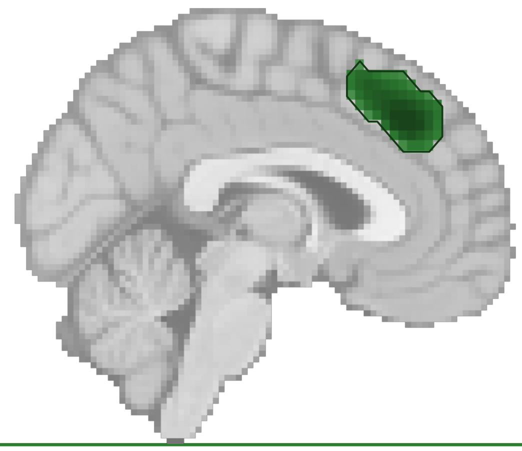
Map not available

▲ Unsigned RPE — Positive

2 connected components (18-conn) · 7 local maxima ≥8 mm apart · 1,450 voxels (2.4% of mask) · 45.7 cm³ · peak |t| = 8.0



#1 35,595 mm³ t=+7.2
y=24 peak (36, 24, -5)

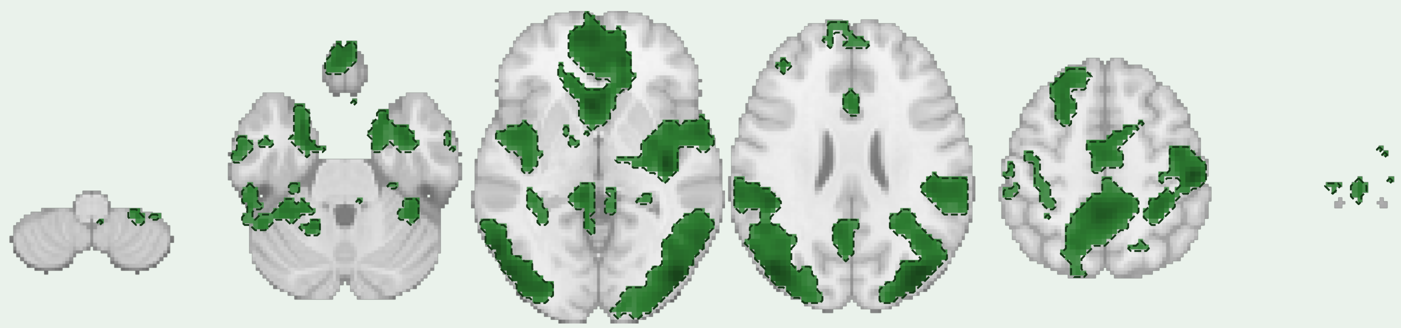


#2 10,080 mm³ t=+8.0
x=2 peak (2, 32, 37)

▼ Unsigned RPE — Negative

One component covers the map — cluster-extent inference is uninformative here

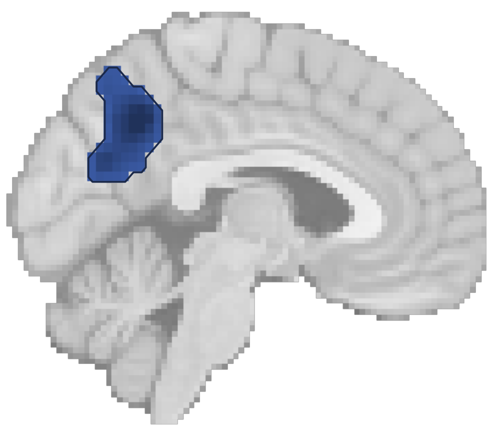
1 connected component (18-conn) · 15 local maxima ≥8 mm apart · 15,194 voxels (25.4% of mask) · 478.6 cm³ · peak |t| = 8.1



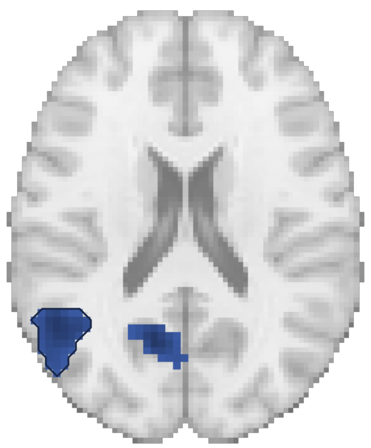
#1 478,611 mm³ t=-8.1
6 axial cuts peak (2, 18, -8)

▲ Shannon surprise — Positive

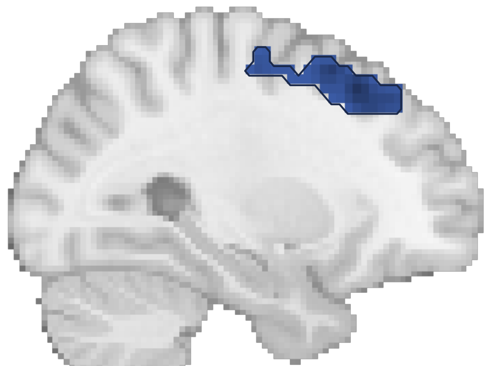
3 connected components (18-conn) · 12 local maxima ≥8 mm apart · 894 voxels (1.5% of mask) · 28.2 cm³ · peak |t| = 6.0



#1 10,490 mm³ t=+6.0
x=-4 peak (-4, -54, 40)



#2 9,891 mm³ t=+4.8
z=20 peak (-52, -54, 20)



#3 7,780 mm³ t=+5.2
x=-28 peak (-28, 24, 48)

▼ Shannon surprise — Negative

Map not available

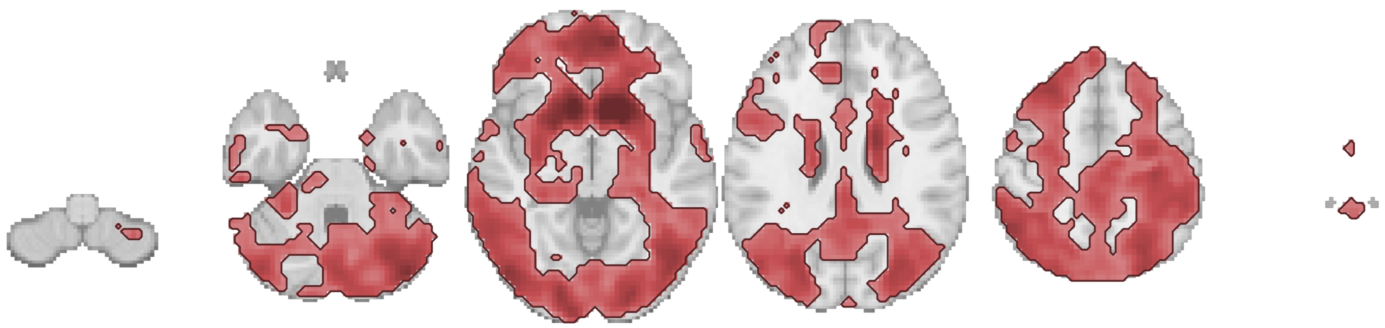
SnPM cluster-extent FWE — model7

Cluster-forming t > 3.2395 (one-tailed p < 0.001, df = 57); clusters with p_FWE < .05, 5000 permutations, one-tailed family per tail. Each slice is cut through one cluster's peak.

▲ Signed RPE — Positive

One component covers the map — cluster-extent inference is uninformative here

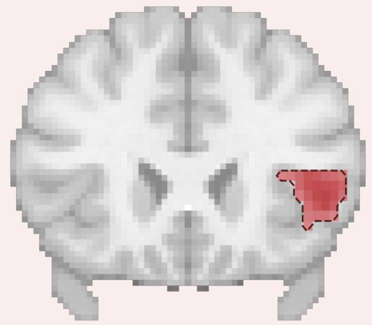
1 connected component (18-conn) · 21 local maxima ≥8 mm apart · 29,988 voxels (50.1% of mask) · 944.6 cm³ · peak |t| = 20.3



#1 944,622 mm³ t=+20.3
6 axial cuts peak (14, 14, -8)

▼ Signed RPE — Negative

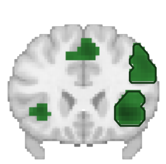
1 connected component (18-conn) · 1 local maxima ≥8 mm apart · 88 voxels (0.1% of mask) · 2.8 cm³ · peak |t| = 6.5



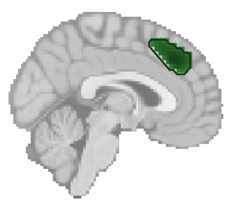
#1 2,772 mm³ t=-6.5
y=24 peak (48, 24, 2)

▲ Unsigned RPE — Positive

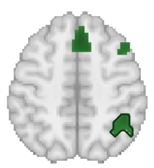
4 connected components (18-conn) · 13 local maxima ≥8 mm apart · 1,215 voxels (2.0% of mask) · 38.3 cm³ · peak |t| = 8.0



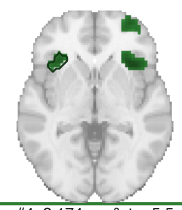
#1 25,546 mm³ t=+7.2
y=24 peak (36, 24, -5)



#2 7,402 mm³ t=+8.0
x=2 peak (2, 32, 37)



#3 3,150 mm³ t=+4.7
z=48 peak (44, -46, 48)

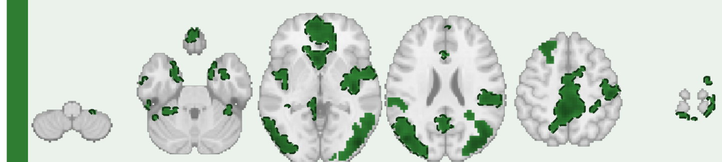


#4 2,174 mm³ t=+5.5
z=-2 peak (-34, 26, -2)

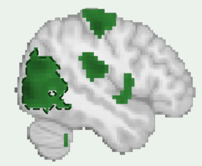
▼ Unsigned RPE — Negative

One component covers the map — cluster-extent inference is uninformative here

4 connected components (18-conn) · 28 local maxima ≥8 mm apart · 10,011 voxels (16.7% of mask) · 315.3 cm³ · peak |t| = 8.1

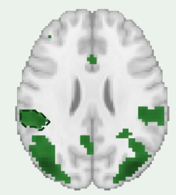


#1 257,292 mm³ t=-8.1
6 axial cuts peak (2, 18, -8)



#2 41,769 mm³ t=-7.3
x=48 peak (48, -82, 2)

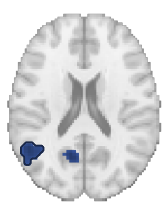
#3 9,576 mm³ t=-6.6
y=32 peak (-34, 32, 40)



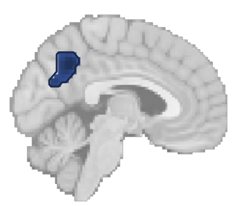
#4 6,710 mm³ t=-5.4
z=26 peak (-48, -36, 26)

▲ Shannon surprise — Positive

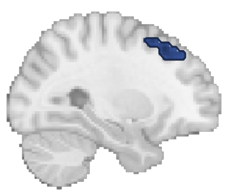
4 connected components (18-conn) · 13 local maxima ≥8 mm apart · 590 voxels (1.0% of mask) · 18.6 cm³ · peak |t| = 6.0



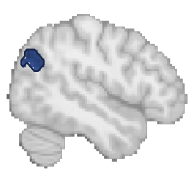
#1 6,741 mm³ t=+4.8
z=20 peak (-52, -54, 20)



#2 5,828 mm³ t=+6.0
x=-4 peak (-4, -54, 40)



#3 3,465 mm³ t=+5.2
x=-28 peak (-28, 24, 48)



#4 2,552 mm³ t=+5.0
x=48 peak (48, -72, 34)

▼ Shannon surprise — Negative

Map not available

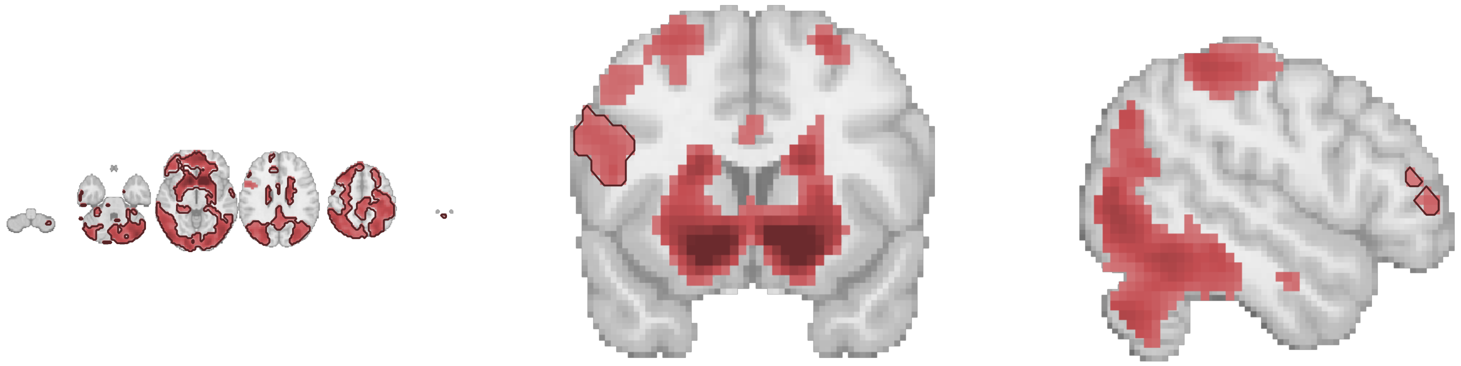
SnPM cluster-extent FWE — model7

Cluster-forming t > 3.9756 (one-tailed p < 0.0001, df = 57); clusters with p_FWE < .05, 5000 permutations, one-tailed family per tail. Each slice is cut through one cluster's peak.

▲ Signed RPE — Positive

One component covers the map — cluster-extent inference is uninformative here

3 connected components (18-conn) · 16 local maxima ≥8 mm apart · 22,421 voxels (37.5% of mask) · 706.3 cm³ · peak |t| = 20.3



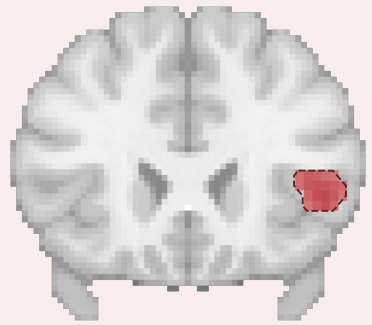
#1 701,978 mm³ t=+20.3
6 axial cuts peak (14, 14, -8)

#2 3,748 mm³ t=+5.7
y=12 peak (-48, 12, 16)

#3 536 mm³ t=+5.1
x=50 peak (50, 44, 6)

▼ Signed RPE — Negative

1 connected component (18-conn) · 1 local maxima ≥8 mm apart · 53 voxels (0.1% of mask) · 1.7 cm³ · peak |t| = 6.5



#1 1,670 mm³ t=-6.5
y=24 peak (48, 24, 2)

▲ Unsigned RPE — Positive

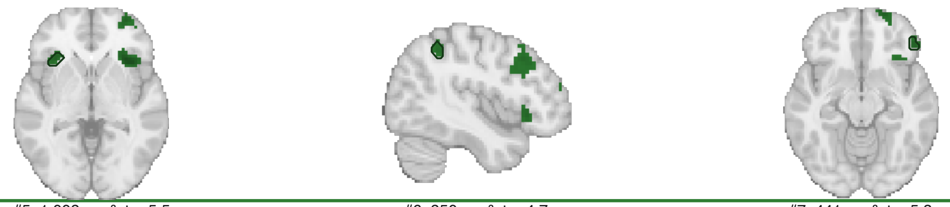
7 connected components (18-conn) · 14 local maxima ≥8 mm apart · 592 voxels (1.0% of mask) · 18.6 cm³ · peak |t| = 8.0

#1 4,504 mm³ t=+8.0
x=2 peak (2, 32, 37)

#2 4,126 mm³ t=+5.0
x=48 peak (48, 24, 37)

#3 4,000 mm³ t=+5.5
y=60 peak (32, 60, 2)

#4 3,717 mm³ t=+7.2
y=24 peak (36, 24, -5)



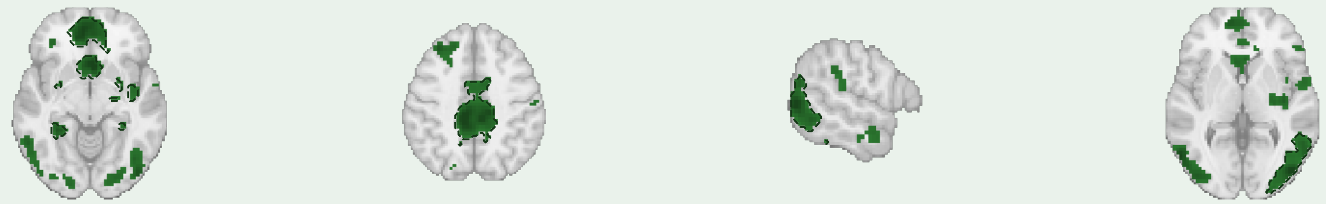
#5 1,008 mm³ t=+5.5
z=-2 peak (-34, 26, -2)

#6 850 mm³ t=+4.7
x=44 peak (44, -46, 48)

#7 441 mm³ t=+5.3
z=-12 peak (50, 36, -12)

▼ Unsigned RPE — Negative

16 connected components (18-conn) · 54 local maxima ≥8 mm apart · 5,137 voxels (8.6% of mask) · 161.8 cm³ · peak |t| = 8.1 · +8 more components (93% of volume shown)

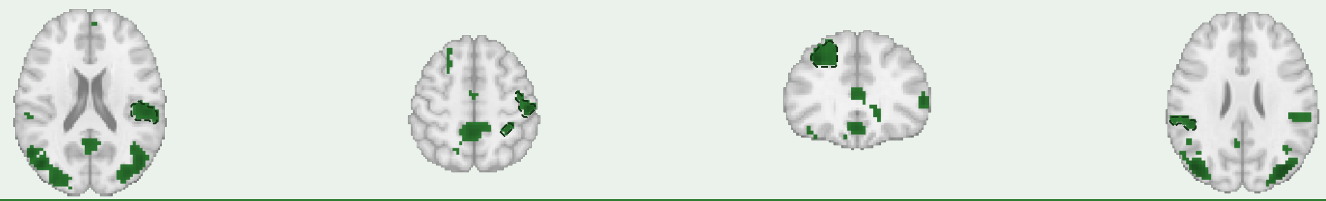


#1 52,920 mm³ t=-8.1
z=-8 peak (2, 18, -8)

#2 28,382 mm³ t=-8.0
z=44 peak (-10, -30, 44)

#3 25,924 mm³ t=-7.4
x=-58 peak (-58, -66, 2)

#4 21,956 mm³ t=-7.3
z=2 peak (48, -82, 2)



#5 6,930 mm³ t=-5.9
z=20 peak (54, -24, 20)

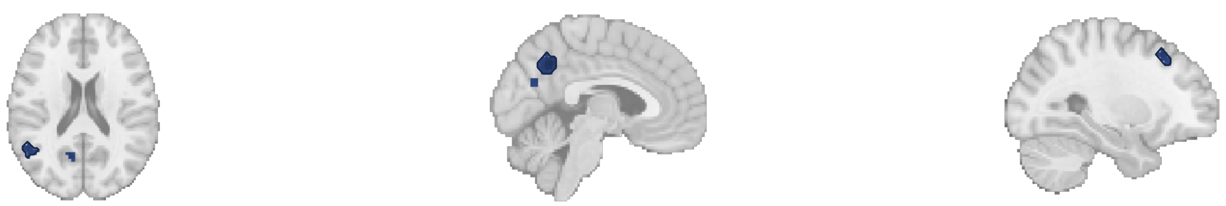
#6 6,772 mm³ t=-6.4
z=54 peak (50, -22, 54)

#7 5,072 mm³ t=-6.6
y=32 peak (-34, 32, 40)

#8 2,835 mm³ t=-5.4
z=26 peak (-48, -36, 26)

▲ Shannon surprise — Positive

5 connected components (18-conn) · 13 local maxima ≥8 mm apart · 211 voxels (0.4% of mask) · 6.6 cm³ · peak |t| = 6.0



#1 2,426 mm³ t=+4.8
z=20 peak (-52, -54, 20)

#2 1,796 mm³ t=+6.0
x=-4 peak (-4, -54, 40)

#3 976 mm³ t=+5.2
x=-28 peak (-28, 24, 48)

#4 850 mm³ t=+5.0
x=48 peak (48, -72, 34)

#5 598 mm³ t=+4.8
z=20 peak (-12, -60, 20)

▼ Shannon surprise — Negative

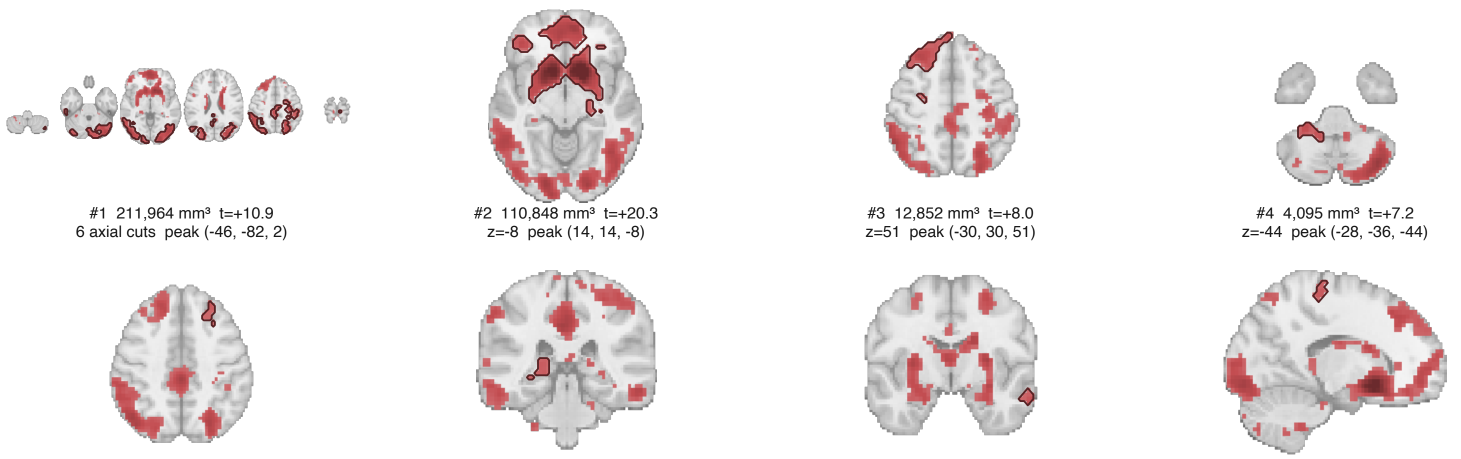
Map not available

SnPM cluster-extent FWE — model7

Cluster-forming t > 5.2929 (one-tailed p < 1e-06, df = 57); clusters with p_FWE < .05, 5000 permutations, one-tailed family per tail. Each slice is cut through one cluster's peak.

▲ Signed RPE — Positive

28 connected components (18-conn) · 65 local maxima ≥8 mm apart · 11,013 voxels (18.4% of mask) · 346.9 cm³ · peak |t| = 20.3 · +20 more components (99% of volume shown)



#1 211,964 mm³ t=+10.9
6 axial cuts peak (-46, -82, 2)

#2 110,848 mm³ t=+20.3
z=-8 peak (14, 14, -8)

#3 12,852 mm³ t=+8.0
z=51 peak (-30, 30, 51)

#4 4,095 mm³ t=+7.2
z=-44 peak (-28, -36, -44)

#5 1,323 mm³ t=+6.4
z=44 peak (26, 38, 44)

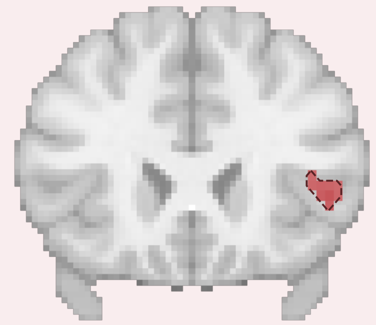
#6 976 mm³ t=+6.1
y=-34 peak (-22, -34, 6)

#7 914 mm³ t=+6.8
y=-6 peak (66, -6, -22)

#8 819 mm³ t=+6.3
x=-16 peak (-16, -28, 65)

▼ Signed RPE — Negative

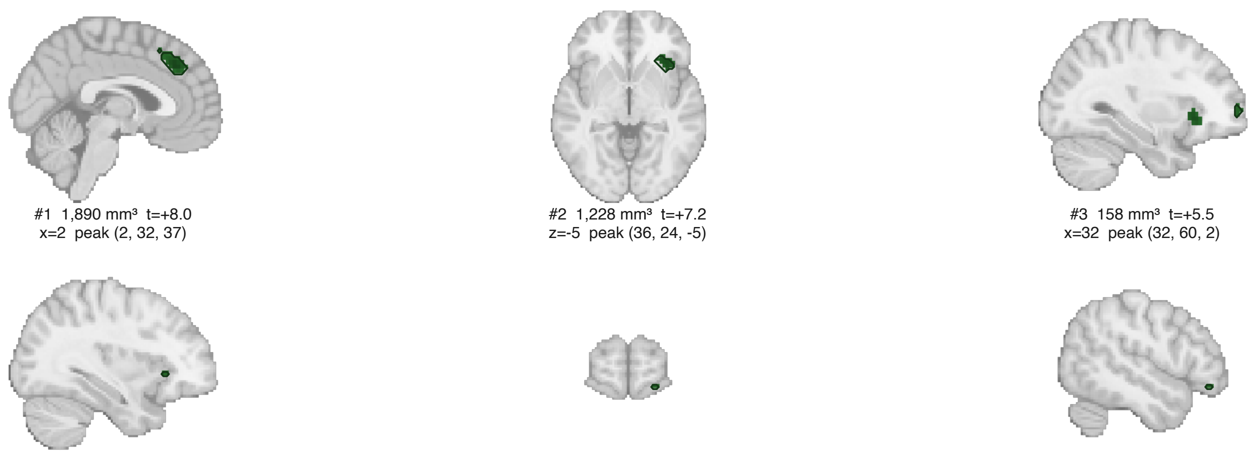
1 connected component (18-conn) · 1 local maxima ≥8 mm apart · 16 voxels (0.0% of mask) · 0.5 cm³ · peak |t| = 6.5



#1 504 mm³ t=-6.5
y=24 peak (48, 24, 2)

▲ Unsigned RPE — Positive

6 connected components (18-conn) · 7 local maxima ≥8 mm apart · 110 voxels (0.2% of mask) · 3.5 cm³ · peak |t| = 8.0



#1 1,890 mm³ t=+8.0
x=2 peak (2, 32, 37)

#2 1,228 mm³ t=+7.2
z=-5 peak (36, 24, -5)

#3 158 mm³ t=+5.5
x=32 peak (32, 60, 2)

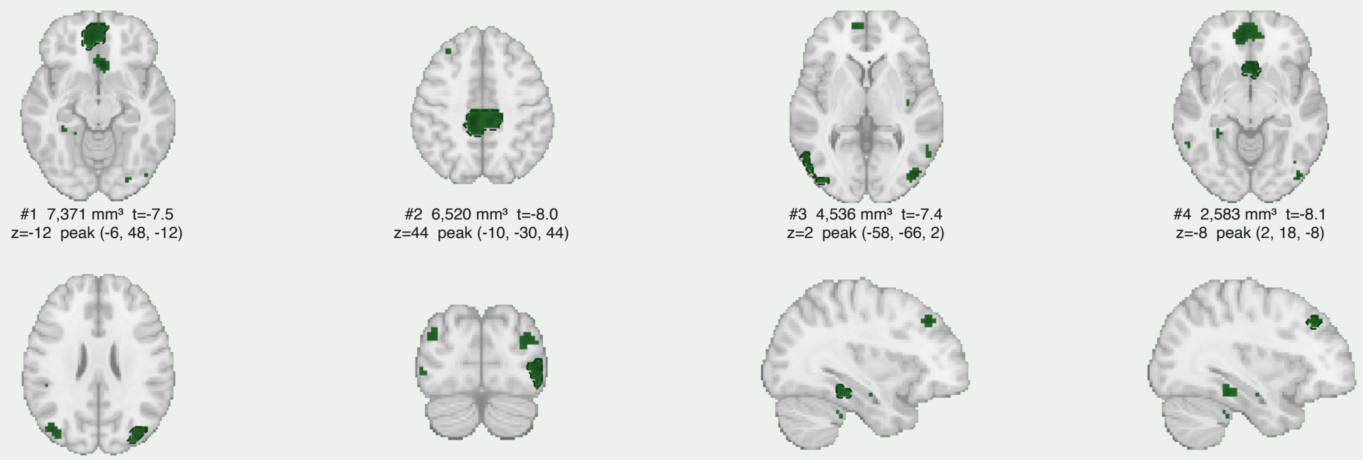
#4 63 mm³ t=+5.5
x=-34 peak (-34, 26, -2)

#5 63 mm³ t=+5.4
y=66 peak (20, 66, -12)

#6 63 mm³ t=+5.3
x=50 peak (50, 36, -12)

▼ Unsigned RPE — Negative

32 connected components (18-conn) · 51 local maxima ≥8 mm apart · 977 voxels (1.6% of mask) · 30.8 cm³ · peak |t| = 8.1 · +24 more components (86% of volume shown)



#1 7,371 mm³ t=-7.5
z=-12 peak (-6, 48, -12)

#2 6,520 mm³ t=-8.0
z=44 peak (-10, -30, 44)

#3 4,536 mm³ t=-7.4
z=2 peak (-58, -66, 2)

#4 2,583 mm³ t=-8.1
z=-8 peak (2, 18, -8)

#5 2,174 mm³ t=-6.8
z=26 peak (38, -78, 26)

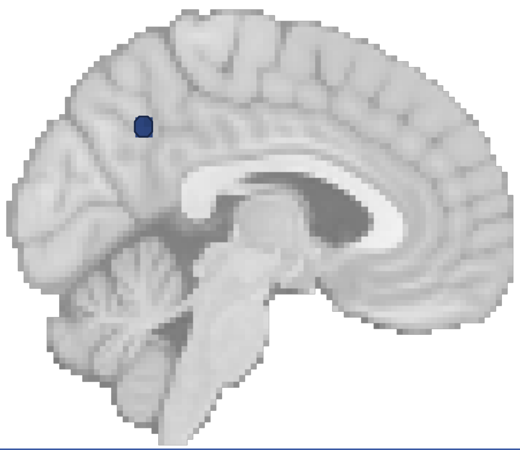
#6 1,890 mm³ t=-7.3
y=-82 peak (48, -82, 2)

#7 850 mm³ t=-6.2
x=-34 peak (-34, -36, -16)

#8 536 mm³ t=-6.6
x=-34 peak (-34, 32, 40)

▲ Shannon surprise — Positive

1 connected component (18-conn) · 1 local maxima ≥8 mm apart · 6 voxels (0.0% of mask) · 0.2 cm³ · peak |t| = 6.0



#1 189 mm³ t=+6.0
x=-4 peak (-4, -54, 40)

▼ Shannon surprise — Negative

Map not available

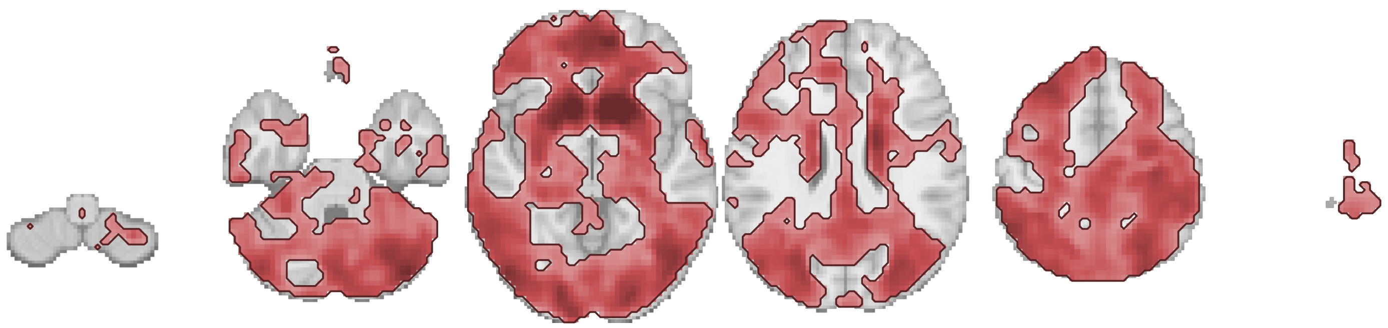
nilearn cluster-extent FWE — model7

Cluster-forming t > 2.3936 (one-tailed p < 0.01); clusters with two-tailed p_FWE < .05, 5000 sign flips, 18-connectivity.

▲ Signed RPE — Positive

One component covers the map — cluster-extent inference is uninformative here

1 connected component (18-conn) · 11 local maxima ≥8 mm apart · 38,852 voxels (64.9% of mask) · 1,223.8 cm³ · peak |t| = 20.3



#1 1,223,838 mm³ t=+20.3
6 axial cuts peak (14, 14, -8)

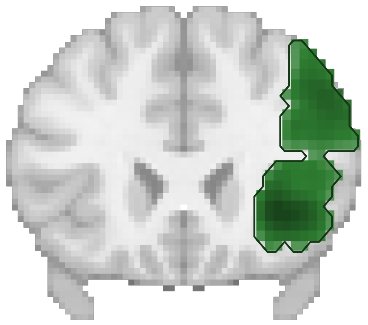
▼ Signed RPE — Negative

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm³ · peak |t| = 0.0

No suprathreshold clusters

▲ Unsigned RPE — Positive

1 connected component (18-conn) · 4 local maxima ≥8 mm apart · 1,299 voxels (2.2% of mask) · 40.9 cm³ · peak |t| = 7.2

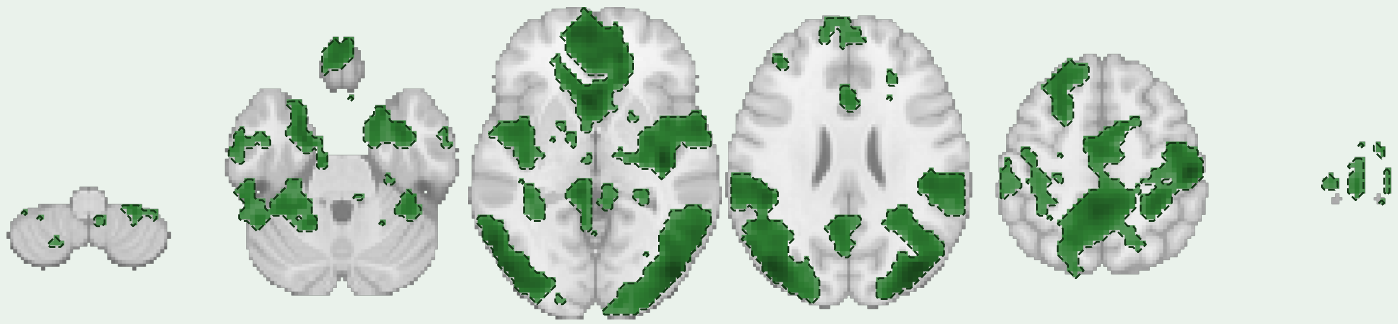


#1 40,918 mm³ t=+7.2
y=24 peak (36, 24, -5)

▼ Unsigned RPE — Negative

One component covers the map — cluster-extent inference is uninformative here

1 connected component (18-conn) · 17 local maxima ≥8 mm apart · 18,112 voxels (30.3% of mask) · 570.5 cm³ · peak |t| = 8.1



#1 570,528 mm³ t=-8.1
6 axial cuts peak (2, 18, -8)

▲ Shannon surprise — Positive

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm³ · peak |t| = 0.0

No suprathreshold clusters

▼ Shannon surprise — Negative

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm³ · peak |t| = 0.0

No suprathreshold clusters

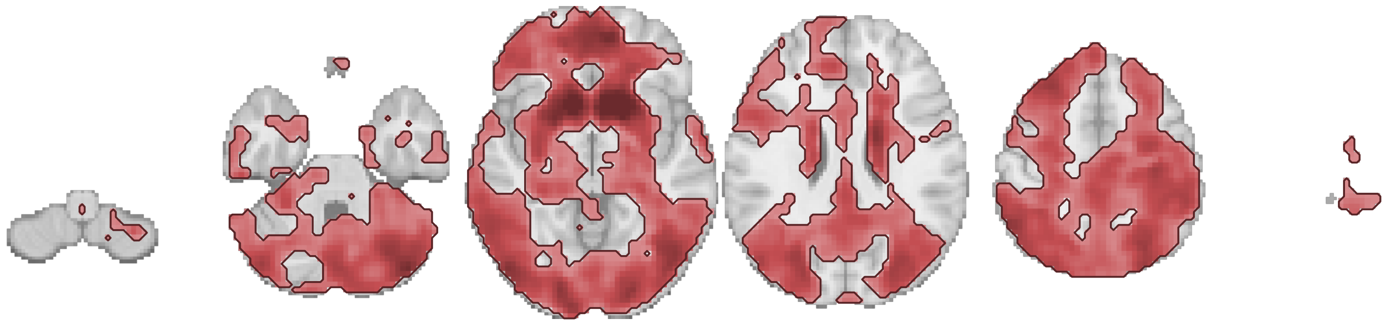
nilearn cluster-extent FWE — model7

Cluster-forming $t > 2.6649$ (one-tailed $p < 0.005$); clusters with two-tailed $p_{FWE} < .05$, 5000 sign flips, 18-connectivity.

▲ Signed RPE — Positive

One component covers the map — cluster-extent inference is uninformative here

1 connected component (18-conn) · 8 local maxima ≥8 mm apart · 36,097 voxels (60.3% of mask) · 1,137.1 cm^3 · peak $|t|$ = 20.3



#1 1,137,056 mm^3 t=+20.3
6 axial cuts peak (14, 14, -8)

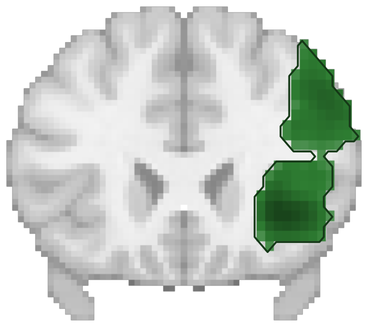
▼ Signed RPE — Negative

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm^3 · peak $|t|$ = 0.0

No suprathreshold clusters

▲ Unsigned RPE — Positive

1 connected component (18-conn) · 4 local maxima ≥8 mm apart · 1,130 voxels (1.9% of mask) · 35.6 cm^3 · peak $|t|$ = 7.2

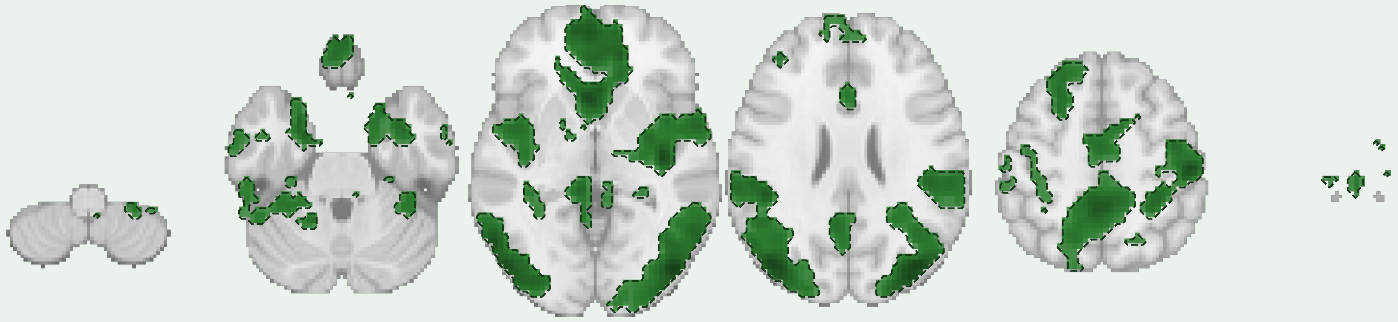


#1 35,595 mm^3 t=+7.2
y=24 peak (36, 24, -5)

▼ Unsigned RPE — Negative

One component covers the map — cluster-extent inference is uninformative here

1 connected component (18-conn) · 15 local maxima ≥8 mm apart · 15,195 voxels (25.4% of mask) · 478.6 cm^3 · peak $|t|$ = 8.1



#1 478,642 mm^3 t=-8.1
6 axial cuts peak (2, 18, -8)

▲ Shannon surprise — Positive

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm^3 · peak $|t|$ = 0.0

No suprathreshold clusters

▼ Shannon surprise — Negative

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm^3 · peak $|t|$ = 0.0

No suprathreshold clusters

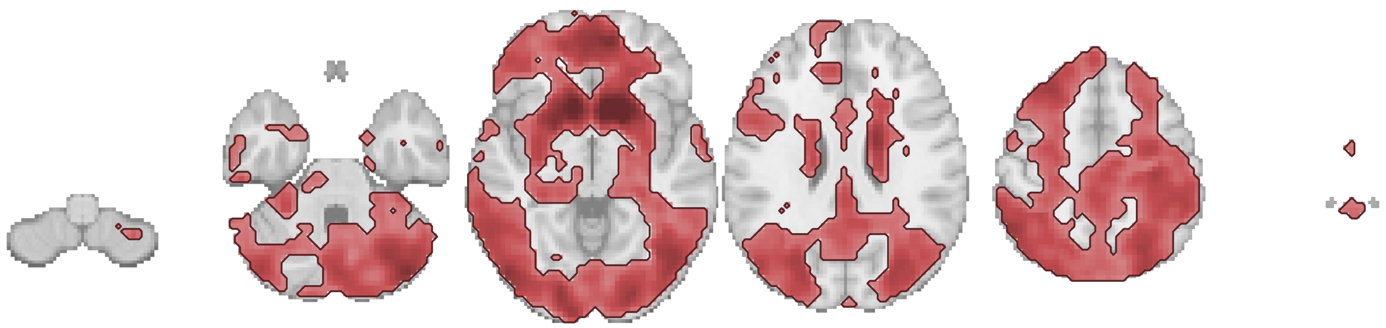
nilearn cluster-extent FWE — model7

Cluster-forming t > 3.2395 (one-tailed p < 0.001); clusters with two-tailed p_FWE < .05, 5000 sign flips, 18-connectivity.

▲ Signed RPE — Positive

One component covers the map — cluster-extent inference is uninformative here

1 connected component (18-conn) · 21 local maxima ≥8 mm apart · 29,988 voxels (50.1% of mask) · 944.6 cm³ · peak |t| = 20.3



#1 944,622 mm³ t=+20.3
6 axial cuts peak (14, 14, -8)

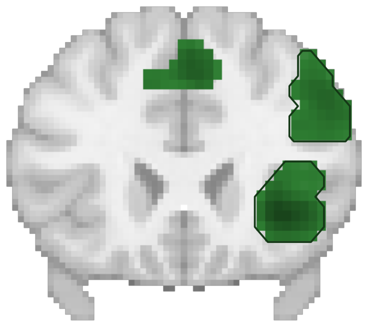
▼ Signed RPE — Negative

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm³ · peak |t| = 0.0

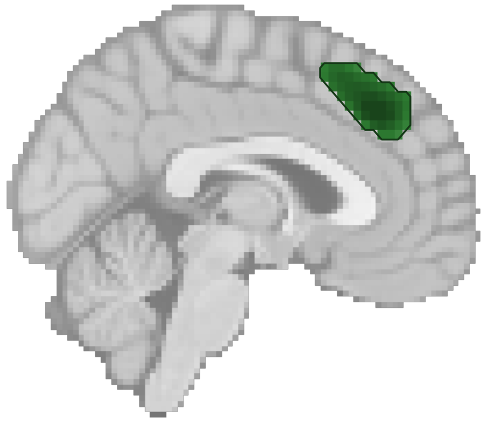
No suprathreshold clusters

▲ Unsigned RPE — Positive

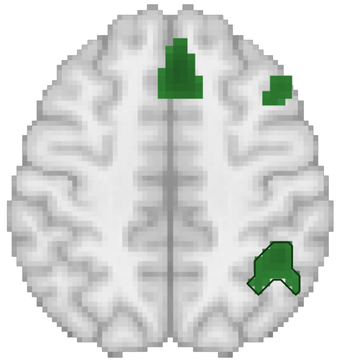
3 connected components (18-conn) · 12 local maxima ≥8 mm apart · 1,146 voxels (1.9% of mask) · 36.1 cm³ · peak |t| = 8.0



#1 25,546 mm³ t=+7.2
y=24 peak (36, 24, -5)



#2 7,402 mm³ t=+8.0
x=2 peak (2, 32, 37)

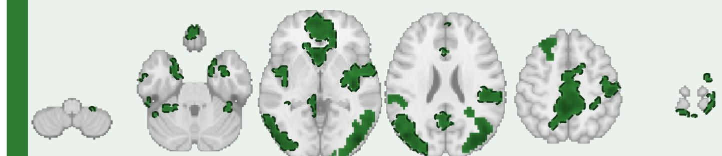


#3 3,150 mm³ t=+4.7
z=48 peak (44, -46, 48)

▼ Unsigned RPE — Negative

One component covers the map — cluster-extent inference is uninformative here

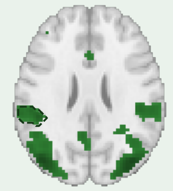
4 connected components (18-conn) · 28 local maxima ≥8 mm apart · 10,011 voxels (16.7% of mask) · 315.3 cm³ · peak |t| = 8.1



#1 257,292 mm³ t=-8.1
6 axial cuts peak (2, 18, -8)

#2 41,769 mm³ t=-7.3
x=48 peak (48, -82, 2)

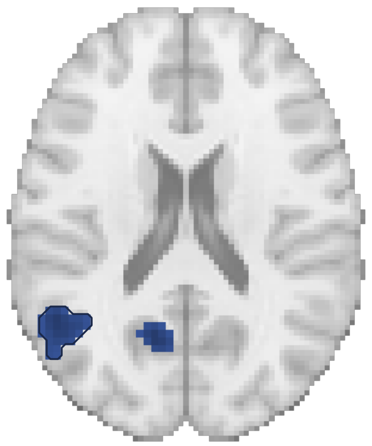
#3 9,576 mm³ t=-6.6
y=32 peak (-34, 32, 40)



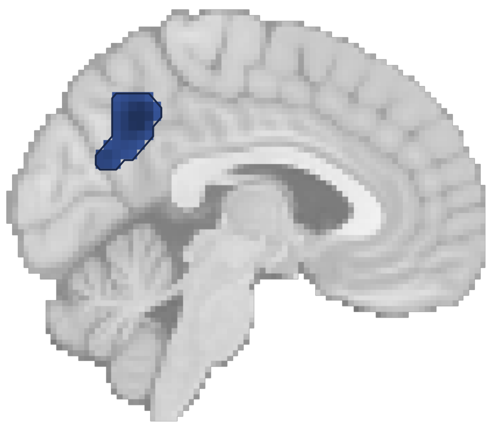
#4 6,710 mm³ t=-5.4
z=26 peak (-48, -36, 26)

▲ Shannon surprise — Positive

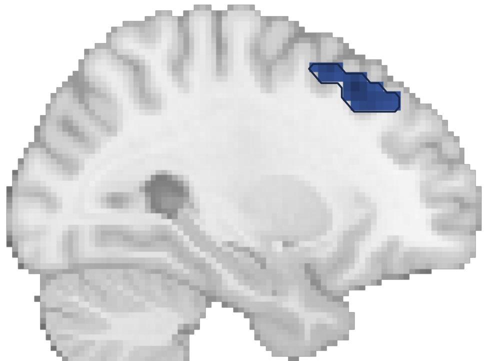
3 connected components (18-conn) · 11 local maxima ≥8 mm apart · 509 voxels (0.9% of mask) · 16.0 cm³ · peak |t| = 6.0



#1 6,741 mm³ t=+4.8
z=20 peak (-52, -54, 20)



#2 5,828 mm³ t=+6.0
x=-4 peak (-4, -54, 40)



#3 3,465 mm³ t=+5.2
x=-28 peak (-28, 24, 48)

▼ Shannon surprise — Negative

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm³ · peak |t| = 0.0

No suprathreshold clusters

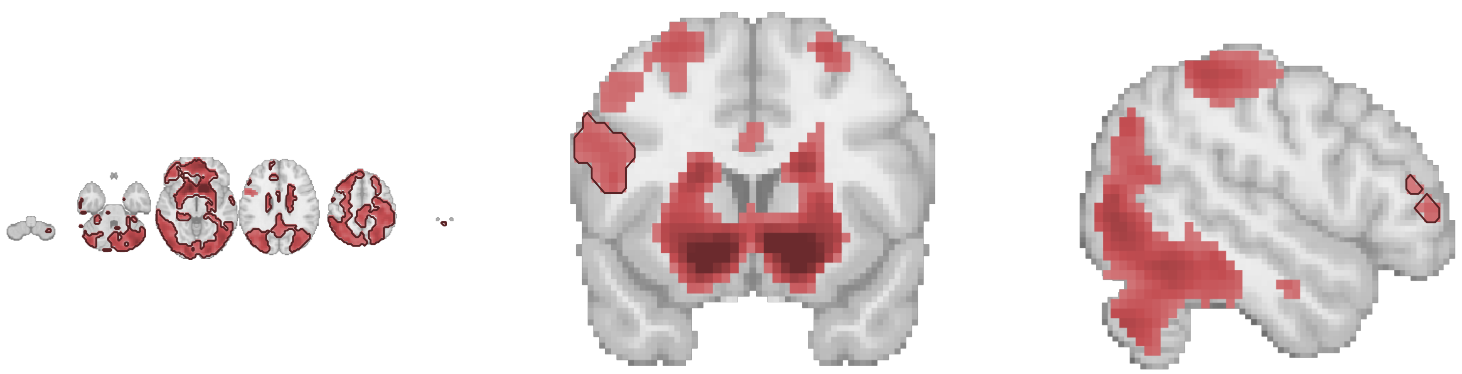
nilearn cluster-extent FWE — model7

Cluster-forming t > 3.9756 (one-tailed p < 0.0001); clusters with two-tailed p_FWE < .05, 5000 sign flips, 18-connectivity.

▲ Signed RPE — Positive

One component covers the map — cluster-extent inference is uninformative here

3 connected components (18-conn) · 16 local maxima ≥8 mm apart · 22,421 voxels (37.5% of mask) · 706.3 cm³ · peak |t| = 20.3



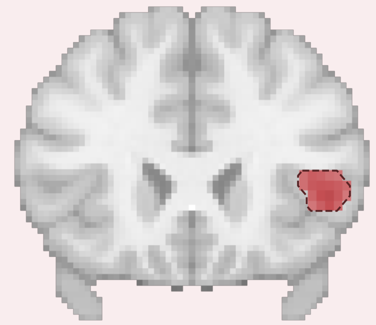
#1 701,978 mm³ t=+20.3
6 axial cuts peak (14, 14, -8)

#2 3,748 mm³ t=+5.7
y=12 peak (-48, 12, 16)

#3 536 mm³ t=+5.1
x=50 peak (50, 44, 6)

▼ Signed RPE — Negative

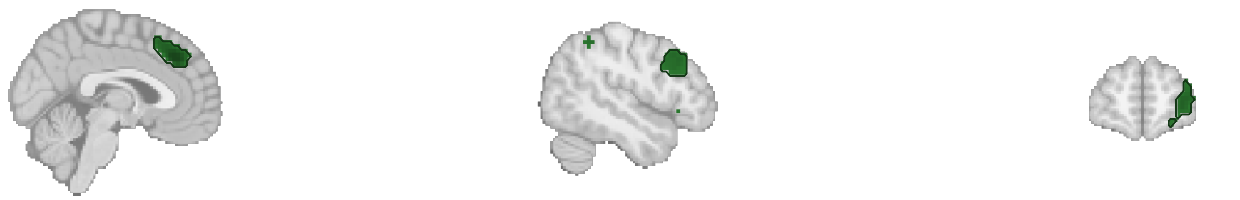
1 connected component (18-conn) · 1 local maxima ≥8 mm apart · 53 voxels (0.1% of mask) · 1.7 cm³ · peak |t| = 6.5



#1 1,670 mm³ t=-6.5
y=24 peak (48, 24, 2)

▲ Unsigned RPE — Positive

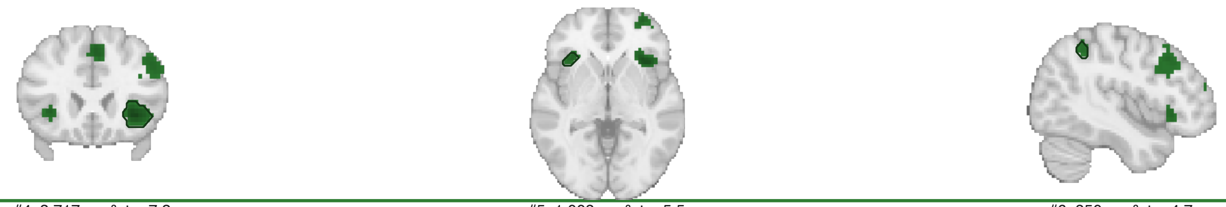
6 connected components (18-conn) · 13 local maxima ≥8 mm apart · 578 voxels (1.0% of mask) · 18.2 cm³ · peak |t| = 8.0



#1 4,504 mm³ t=+8.0
x=2 peak (2, 32, 37)

#2 4,126 mm³ t=+5.0
x=48 peak (48, 24, 37)

#3 4,000 mm³ t=+5.5
y=60 peak (32, 60, 2)



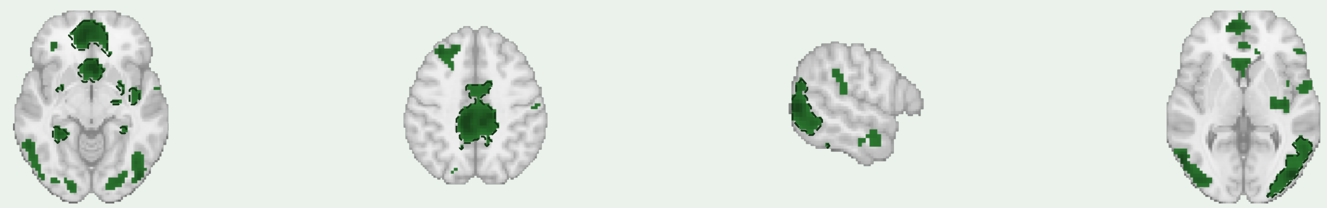
#4 3,717 mm³ t=+7.2
y=24 peak (36, 24, -5)

#5 1,008 mm³ t=+5.5
z=-2 peak (-34, 26, -2)

#6 850 mm³ t=+4.7
x=44 peak (44, -46, 48)

▼ Unsigned RPE — Negative

15 connected components (18-conn) · 53 local maxima ≥8 mm apart · 5,124 voxels (8.6% of mask) · 161.4 cm³ · peak |t| = 8.1 +7 more components (93% of volume shown)

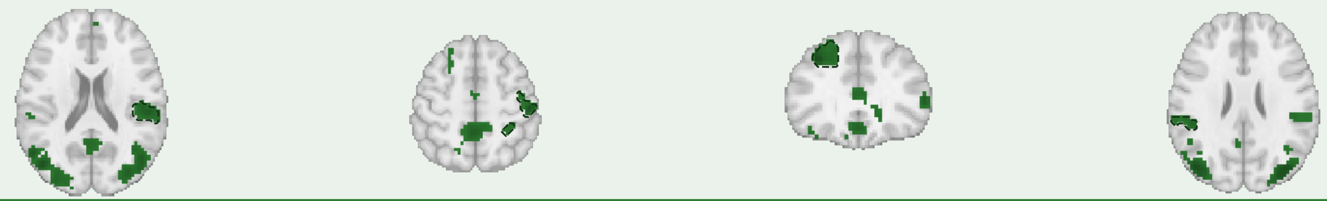


#1 52,920 mm³ t=-8.1
z=-8 peak (2, 18, -8)

#2 28,382 mm³ t=-8.0
z=44 peak (-10, -30, 44)

#3 25,924 mm³ t=-7.4
x=-58 peak (-58, -66, 2)

#4 21,956 mm³ t=-7.3
z=2 peak (48, -82, 2)



#5 6,898 mm³ t=-5.9
z=20 peak (54, -24, 20)

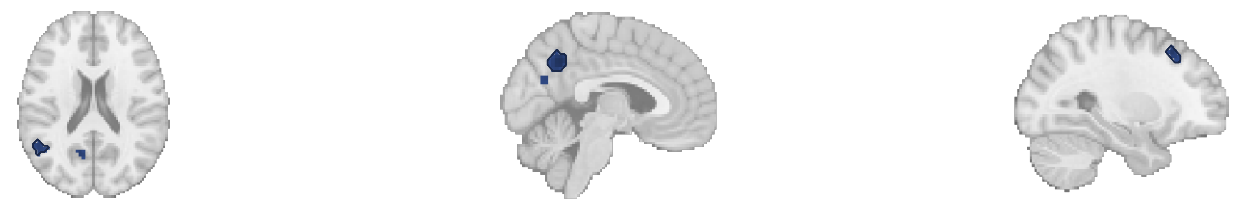
#6 6,772 mm³ t=-6.4
z=54 peak (50, -22, 54)

#7 5,072 mm³ t=-6.6
y=32 peak (-34, 32, 40)

#8 2,835 mm³ t=-5.4
z=26 peak (-48, -36, 26)

▲ Shannon surprise — Positive

5 connected components (18-conn) · 13 local maxima ≥8 mm apart · 210 voxels (0.4% of mask) · 6.6 cm³ · peak |t| = 6.0



#1 2,426 mm³ t=+4.8
z=20 peak (-52, -54, 20)

#2 1,764 mm³ t=+6.0
x=-4 peak (-4, -54, 40)

#3 976 mm³ t=+5.2
x=-28 peak (-28, 24, 48)

#4 850 mm³ t=+5.0
x=48 peak (48, -72, 34)

#5 598 mm³ t=+4.8
z=20 peak (-12, -60, 20)

▼ Shannon surprise — Negative

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm³ · peak |t| = 0.0

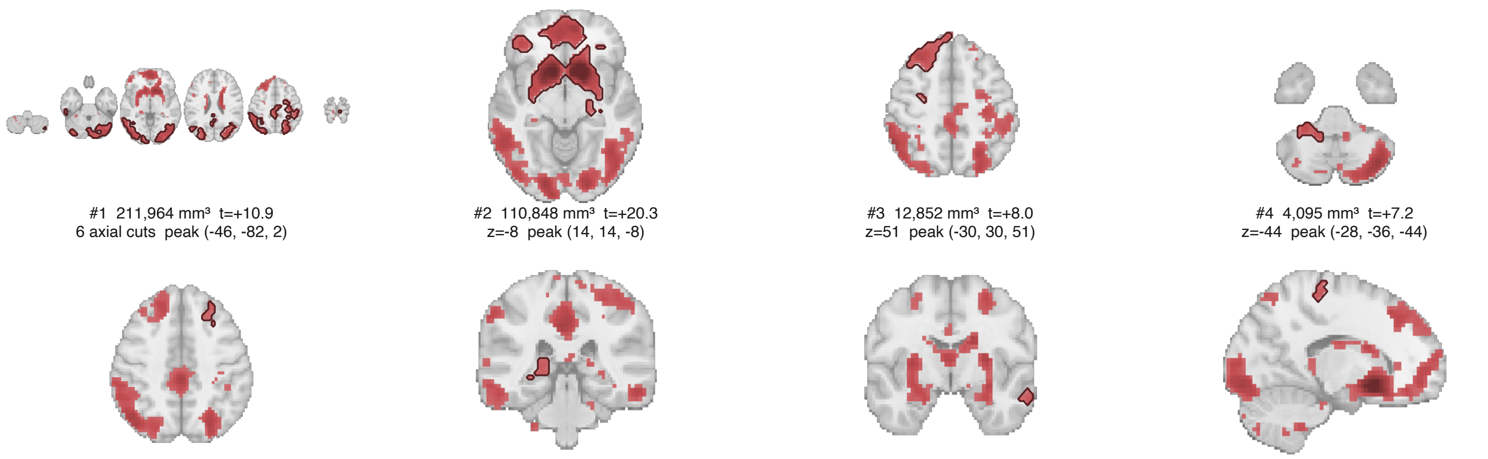
No suprathreshold clusters

nilearn cluster-extent FWE — model7

Cluster-forming t > 5.2929 (one-tailed p < 1e-06); clusters with two-tailed p_FWE < .05, 5000 sign flips, 18-connectivity.

▲ Signed RPE — Positive

28 connected components (18-conn) · 65 local maxima ≥8 mm apart · 11,013 voxels (18.4% of mask) · 346.9 cm³ · peak |t| = 20.3 +20 more components (99% of volume shown)



#1 211,964 mm³ t=+10.9
6 axial cuts peak (-46, -82, 2)

#2 110,848 mm³ t=+20.3
z=-8 peak (14, 14, -8)

#3 12,852 mm³ t=+8.0
z=51 peak (-30, 30, 51)

#4 4,095 mm³ t=+7.2
z=-44 peak (-28, -36, -44)

#5 1,323 mm³ t=+6.4
z=44 peak (26, 38, 44)

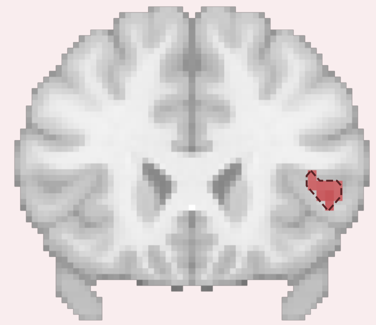
#6 976 mm³ t=+6.1
y=-34 peak (-22, -34, 6)

#7 914 mm³ t=+6.8
y=-6 peak (66, -6, -22)

#8 819 mm³ t=+6.3
x=-16 peak (-16, -28, 65)

▼ Signed RPE — Negative

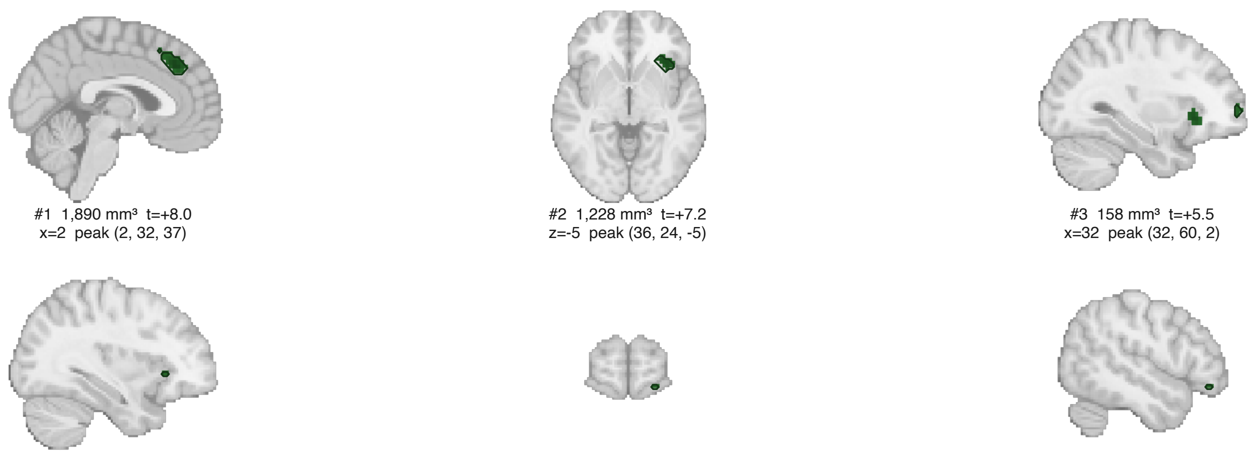
1 connected component (18-conn) · 1 local maxima ≥8 mm apart · 16 voxels (0.0% of mask) · 0.5 cm³ · peak |t| = 6.5



#1 504 mm³ t=-6.5
y=24 peak (48, 24, 2)

▲ Unsigned RPE — Positive

6 connected components (18-conn) · 7 local maxima ≥8 mm apart · 110 voxels (0.2% of mask) · 3.5 cm³ · peak |t| = 8.0



#1 1,890 mm³ t=+8.0
x=2 peak (2, 32, 37)

#2 1,228 mm³ t=+7.2
z=-5 peak (36, 24, -5)

#3 158 mm³ t=+5.5
x=32 peak (32, 60, 2)

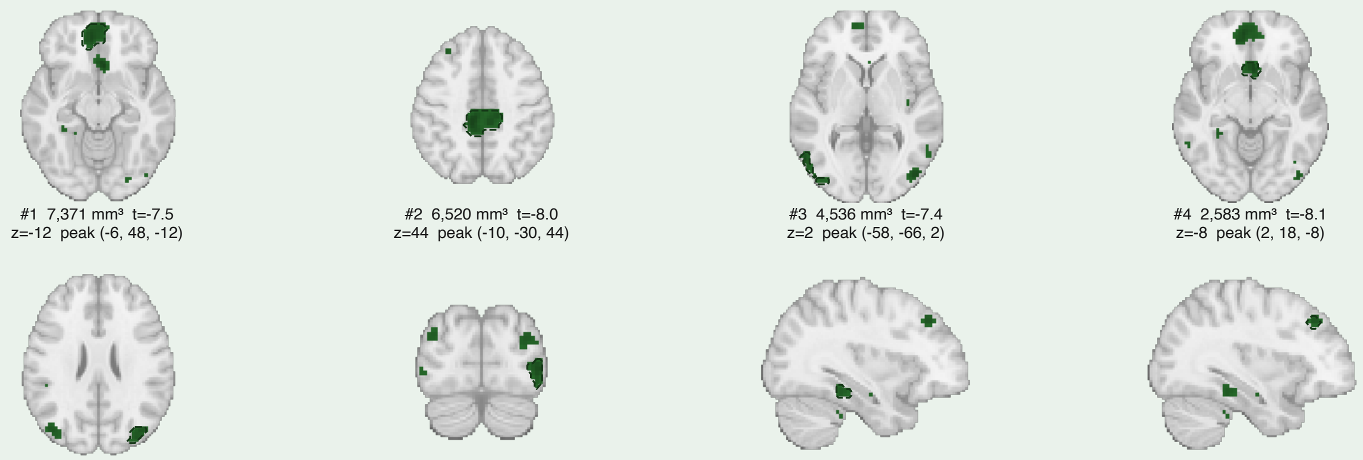
#4 63 mm³ t=+5.5
x=-34 peak (-34, 26, -2)

#5 63 mm³ t=+5.4
y=66 peak (20, 66, -12)

#6 63 mm³ t=+5.3
x=50 peak (50, 36, -12)

▼ Unsigned RPE — Negative

32 connected components (18-conn) · 51 local maxima ≥8 mm apart · 977 voxels (1.6% of mask) · 30.8 cm³ · peak |t| = 8.1 +24 more components (86% of volume shown)



#1 7,371 mm³ t=-7.5
z=-12 peak (-6, 48, -12)

#2 6,520 mm³ t=-8.0
z=44 peak (-10, -30, 44)

#3 4,536 mm³ t=-7.4
z=2 peak (-58, -66, 2)

#4 2,583 mm³ t=-8.1
z=-8 peak (2, 18, -8)

#5 2,174 mm³ t=-6.8
z=26 peak (38, -78, 26)

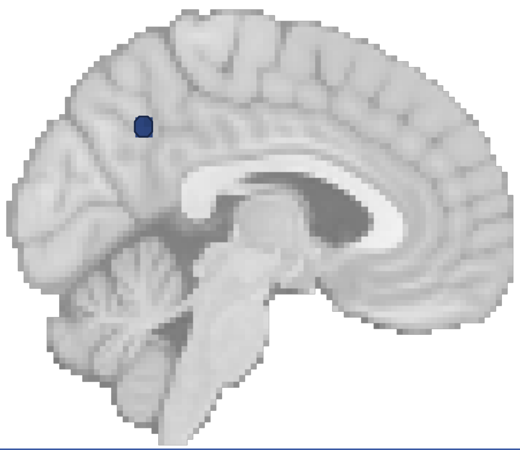
#6 1,890 mm³ t=-7.3
y=-82 peak (48, -82, 2)

#7 850 mm³ t=-6.2
x=-34 peak (-34, -36, -16)

#8 536 mm³ t=-6.6
x=-34 peak (-34, 32, 40)

▲ Shannon surprise — Positive

1 connected component (18-conn) · 1 local maxima ≥8 mm apart · 6 voxels (0.0% of mask) · 0.2 cm³ · peak |t| = 6.0



#1 189 mm³ t=+6.0
x=-4 peak (-4, -54, 40)

▼ Shannon surprise — Negative

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm³ · peak |t| = 0.0

No suprathreshold clusters

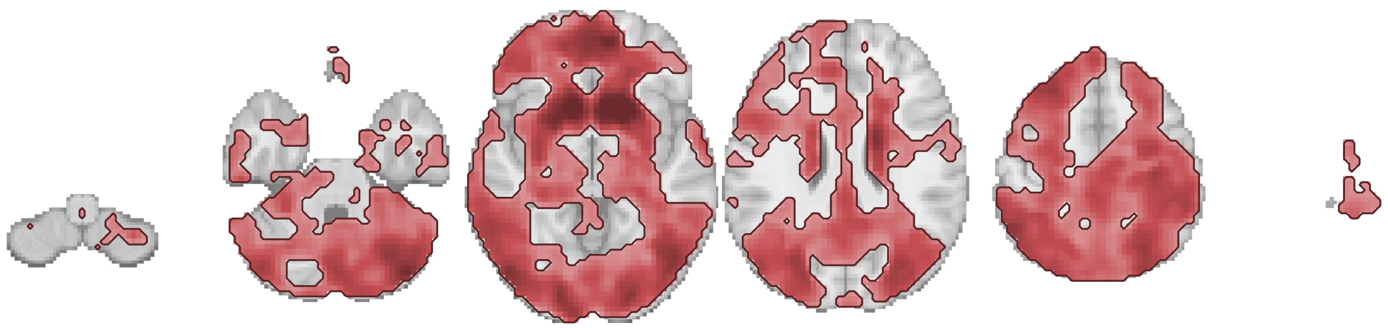
nilearn cluster-MASS FWE — model7

Cluster-forming t > 2.3936 (one-tailed p < 0.01); clusters with two-tailed p_FWE < .05 on summed excess t rather than extent.

▲ Signed RPE — Positive

One component covers the map — cluster-extent inference is uninformative here

1 connected component (18-conn) · 11 local maxima ≥8 mm apart · 38,852 voxels (64.9% of mask) · 1,223.8 cm³ · peak |t| = 20.3



#1 1,223,838 mm³ t=+20.3
6 axial cuts peak (14, 14, -8)

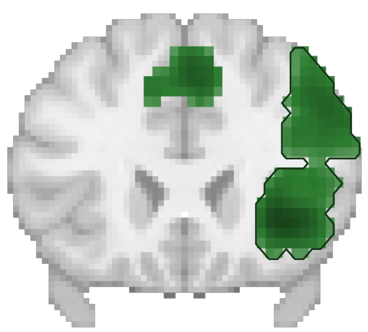
▼ Signed RPE — Negative

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm³ · peak |t| = 0.0

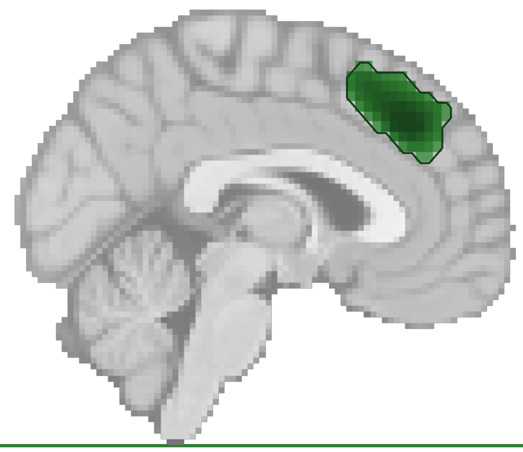
No suprathreshold clusters

▲ Unsigned RPE — Positive

2 connected components (18-conn) · 7 local maxima ≥8 mm apart · 1,668 voxels (2.8% of mask) · 52.5 cm³ · peak |t| = 8.0



#1 40,918 mm³ t=+7.2
y=24 peak (36, 24, -5)

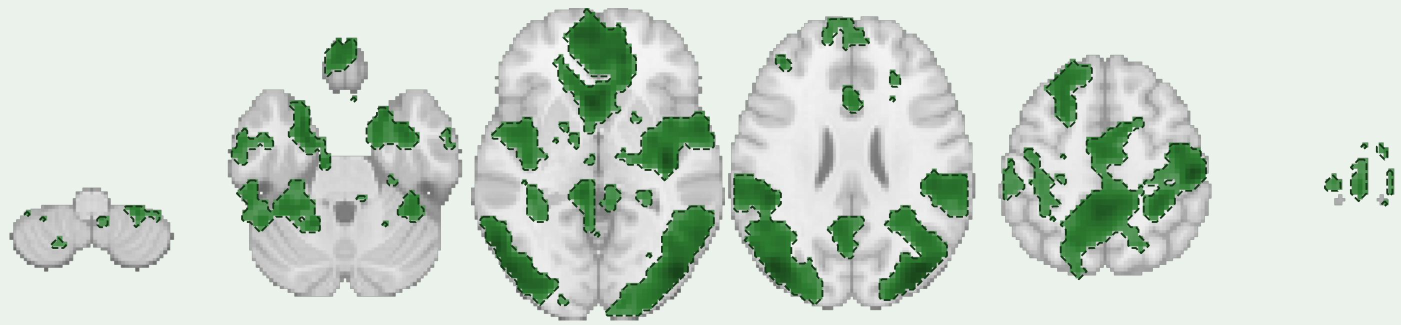


#2 11,624 mm³ t=+8.0
x=2 peak (2, 32, 37)

▼ Unsigned RPE — Negative

One component covers the map — cluster-extent inference is uninformative here

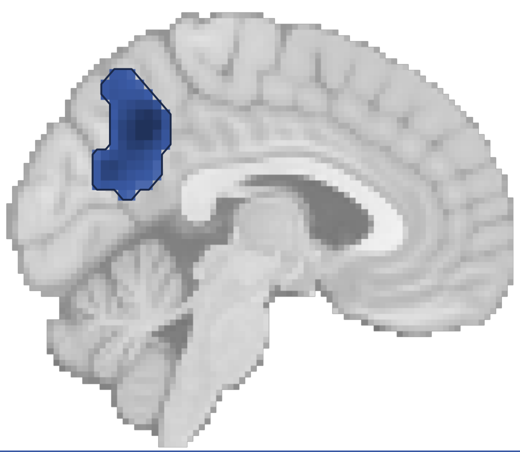
1 connected component (18-conn) · 17 local maxima ≥8 mm apart · 18,112 voxels (30.3% of mask) · 570.5 cm³ · peak |t| = 8.1



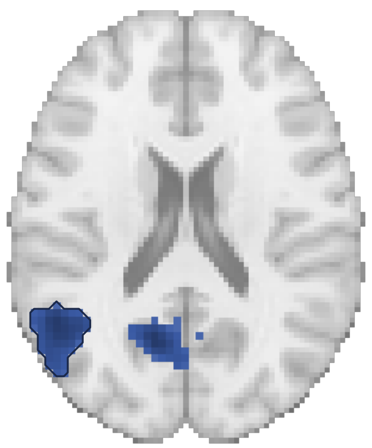
#1 570,528 mm³ t=-8.1
6 axial cuts peak (2, 18, -8)

▲ Shannon surprise — Positive

2 connected components (18-conn) · 8 local maxima ≥8 mm apart · 773 voxels (1.3% of mask) · 24.3 cm³ · peak |t| = 6.0



#1 12,978 mm³ t=+6.0
x=-4 peak (-4, -54, 40)



#2 11,372 mm³ t=+4.8
z=20 peak (-52, -54, 20)

▼ Shannon surprise — Negative

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm³ · peak |t| = 0.0

No suprathreshold clusters

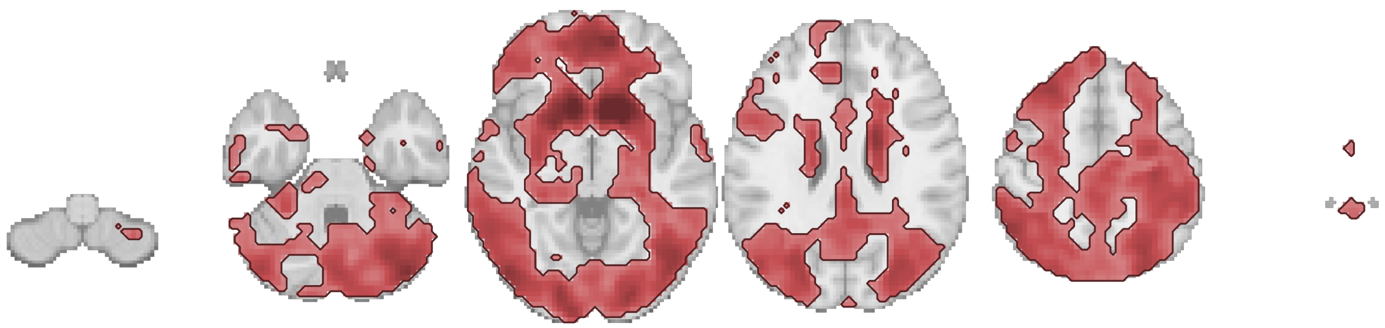
nilearn cluster-MASS FWE — model7

Cluster-forming t > 3.2395 (one-tailed p < 0.001); clusters with two-tailed p_FWE < .05 on summed excess t rather than extent.

▲ Signed RPE — Positive

One component covers the map — cluster-extent inference is uninformative here

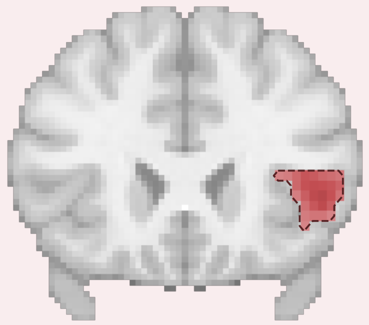
1 connected component (18-conn) · 21 local maxima ≥8 mm apart · 29,988 voxels (50.1% of mask) · 944.6 cm^3 · peak |t| = 20.3



#1 944,622 mm^3 t=+20.3
6 axial cuts peak (14, 14, -8)

▼ Signed RPE — Negative

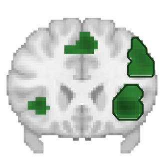
1 connected component (18-conn) · 1 local maxima ≥8 mm apart · 88 voxels (0.1% of mask) · 2.8 cm^3 · peak |t| = 6.5



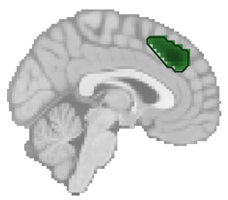
#1 2,772 mm^3 t=-6.5
y=24 peak (48, 24, 2)

▲ Unsigned RPE — Positive

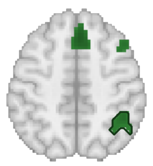
4 connected components (18-conn) · 13 local maxima ≥8 mm apart · 1,215 voxels (2.0% of mask) · 38.3 cm^3 · peak |t| = 8.0



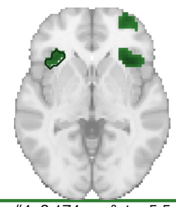
#1 25,546 mm^3 t=+7.2
y=24 peak (36, 24, -5)



#2 7,402 mm^3 t=+8.0
x=2 peak (2, 32, 37)



#3 3,150 mm^3 t=+4.7
z=48 peak (44, -46, 48)

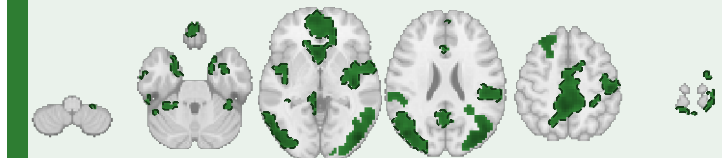


#4 2,174 mm^3 t=+5.5
z=-2 peak (-34, 26, -2)

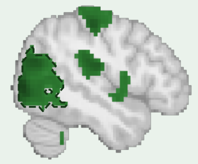
▼ Unsigned RPE — Negative

One component covers the map — cluster-extent inference is uninformative here

4 connected components (18-conn) · 28 local maxima ≥8 mm apart · 10,011 voxels (16.7% of mask) · 315.3 cm^3 · peak |t| = 8.1

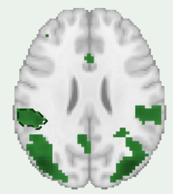


#1 257,292 mm^3 t=-8.1
6 axial cuts peak (2, 18, -8)



#2 41,769 mm^3 t=-7.3
x=48 peak (48, -82, 2)

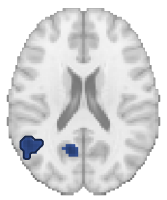
#3 9,576 mm^3 t=-6.6
y=32 peak (-34, 32, 40)



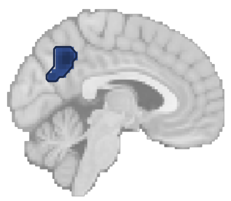
#4 6,710 mm^3 t=-5.4
z=26 peak (-48, -36, 26)

▲ Shannon surprise — Positive

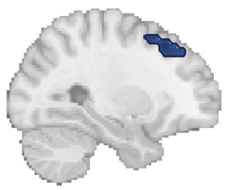
4 connected components (18-conn) · 13 local maxima ≥8 mm apart · 590 voxels (1.0% of mask) · 18.6 cm^3 · peak |t| = 6.0



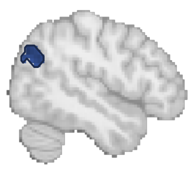
#1 6,741 mm^3 t=+4.8
z=20 peak (-52, -54, 20)



#2 5,828 mm^3 t=+6.0
x=-4 peak (-4, -54, 40)



#3 3,465 mm^3 t=+5.2
x=-28 peak (-28, 24, 48)



#4 2,552 mm^3 t=+5.0
x=48 peak (48, -72, 34)

▼ Shannon surprise — Negative

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm^3 · peak |t| = 0.0

No suprathreshold clusters

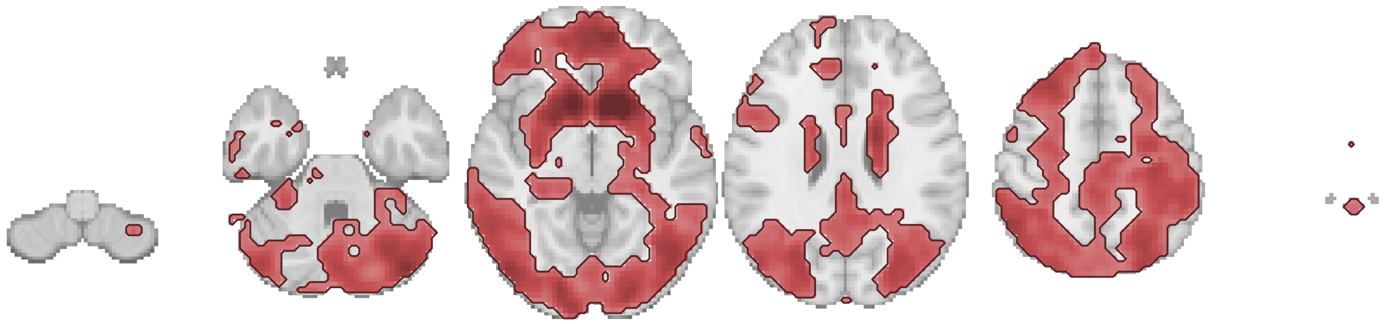
TFCE — model7

Threshold-free cluster enhancement, two-tailed p_FWE < .05, 5000 sign flips. No cluster-forming threshold is chosen at all, so this page is the one that does not depend on the arbitrary decision the rest of the document sweeps over.

▲ Signed RPE — Positive

One component covers the map — cluster-extent inference is uninformative here

1 connected component (18-conn) · 17 local maxima ≥8 mm apart · 25,696 voxels (42.9% of mask) · 809.4 cm³ · peak |t| = 20.3



#1 809,424 mm³ t=+20.3
6 axial cuts peak (14, 14, -8)

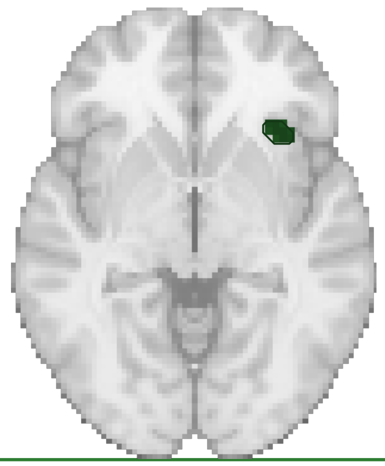
▼ Signed RPE — Negative

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm³ · peak |t| = 0.0

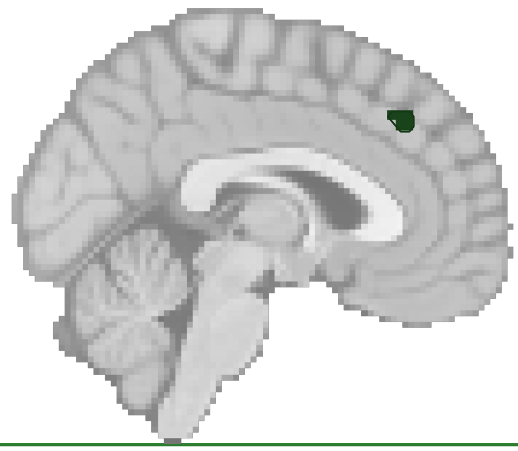
No suprathreshold clusters

▲ Unsigned RPE — Positive

2 connected components (18-conn) · 2 local maxima ≥8 mm apart · 27 voxels (0.0% of mask) · 0.9 cm³ · peak |t| = 8.0



#1 536 mm³ t=+7.2
z=-5 peak (36, 24, -5)

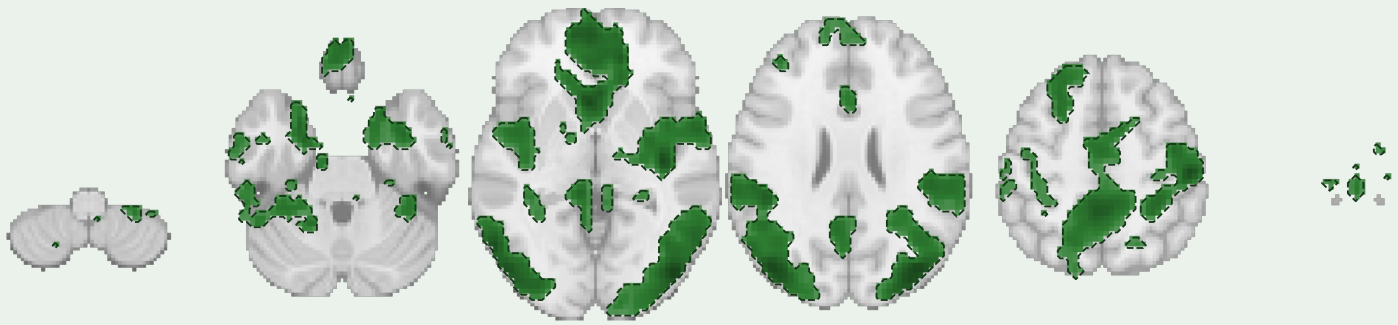


#2 315 mm³ t=+8.0
x=2 peak (2, 32, 37)

▼ Unsigned RPE — Negative

One component covers the map — cluster-extent inference is uninformative here

1 connected component (18-conn) · 10 local maxima ≥8 mm apart · 15,892 voxels (26.6% of mask) · 500.6 cm³ · peak |t| = 8.1



#1 500,598 mm³ t=-8.1
6 axial cuts peak (2, 18, -8)

▲ Shannon surprise — Positive

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm³ · peak |t| = 0.0

No suprathreshold clusters

▼ Shannon surprise — Negative

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm³ · peak |t| = 0.0

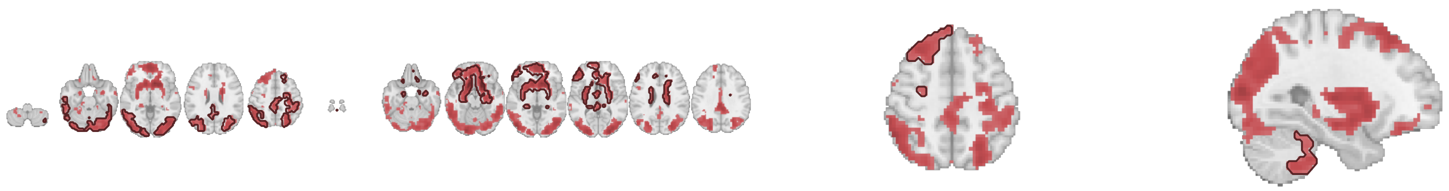
No suprathreshold clusters

Voxel-wise (max-t) FWE — model7

Voxel-level familywise error, two-tailed p_FWE < .05, 5000 sign flips. The strictest and least assumption-laden test here: no spatial criterion of any kind.

▲ Signed RPE — Positive

25 connected components (18-conn) · 55 local maxima ≥8 mm apart · 13,276 voxels (22.2% of mask) · 418.2 cm³ · peak |t| = 20.3 · +17 more components (100% of volume shown)



#1 256,064 mm³ t=+10.9
6 axial cuts peak (-46, -82, 2)

#2 131,544 mm³ t=+20.3
6 axial cuts peak (14, 14, -8)

#3 16,726 mm³ t=+8.0
z=51 peak (-30, 30, 51)

#4 6,584 mm³ t=+7.2
x=-28 peak (-28, -36, -44)

#5 1,796 mm³ t=+6.3
y=-28 peak (-16, -28, 65)

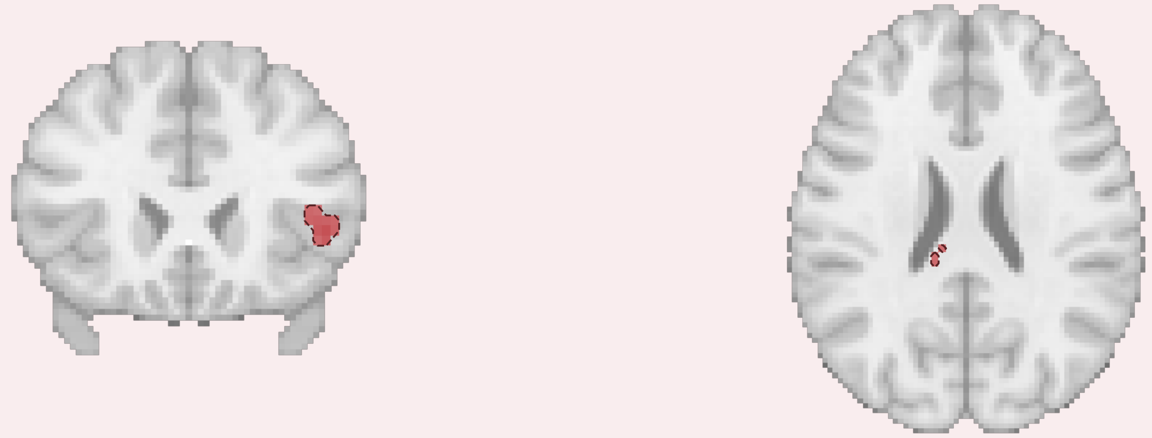
#6 1,228 mm³ t=+6.3
z=-44 peak (8, -46, -44)

#7 1,166 mm³ t=+6.8
z=-22 peak (66, -6, -22)

#8 1,071 mm³ t=+5.7
y=12 peak (-48, 12, 16)

▼ Signed RPE — Negative

2 connected components (18-conn) · 3 local maxima ≥8 mm apart · 27 voxels (0.0% of mask) · 0.9 cm³ · peak |t| = 6.5



#1 756 mm³ t=-6.5
y=24 peak (48, 24, 2)

#2 94 mm³ t=-5.3
z=23 peak (-12, -34, 23)

▲ Unsigned RPE — Positive

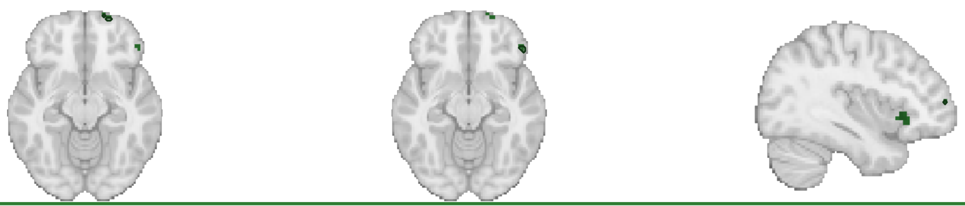
7 connected components (18-conn) · 8 local maxima ≥8 mm apart · 136 voxels (0.2% of mask) · 4.3 cm³ · peak |t| = 8.0

#1 2,205 mm³ t=+8.0
x=2 peak (2, 32, 37)

#2 1,512 mm³ t=+7.2
z=-5 peak (36, 24, -5)

#3 189 mm³ t=+5.5
y=26 peak (-34, 26, -2)

#4 158 mm³ t=+5.5
x=32 peak (32, 60, 2)



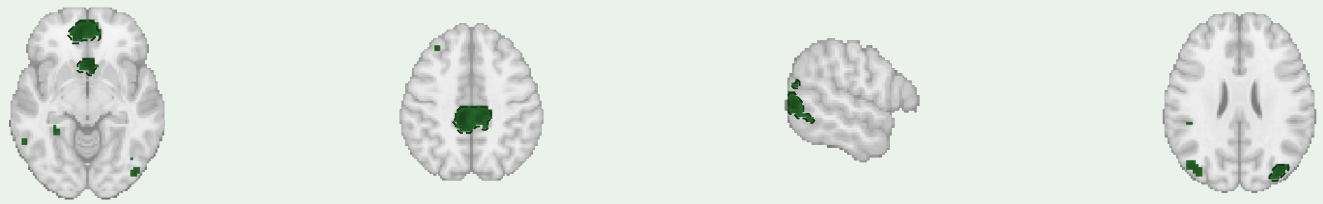
#5 94 mm³ t=+5.4
z=-12 peak (20, 66, -12)

#6 94 mm³ t=+5.3
z=-12 peak (50, 36, -12)

#7 32 mm³ t=+5.2
x=38 peak (38, 56, 9)

▼ Unsigned RPE — Negative

38 connected components (18-conn) · 59 local maxima ≥8 mm apart · 1,198 voxels (2.0% of mask) · 37.7 cm³ · peak |t| = 8.1 · +30 more components (85% of volume shown)

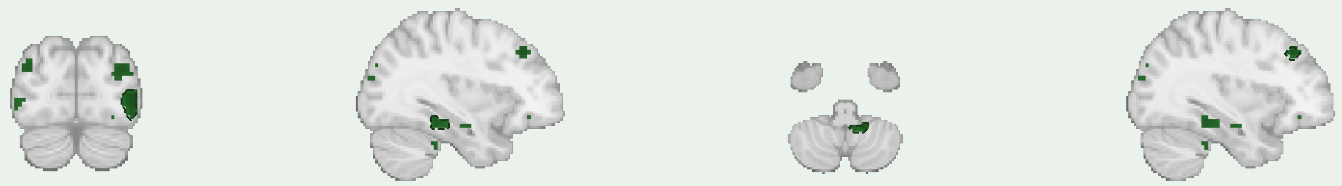


#1 11,214 mm³ t=-8.1
z=-8 peak (2, 18, -8)

#2 7,592 mm³ t=-8.0
z=44 peak (-10, -30, 44)

#3 5,733 mm³ t=-7.4
x=-58 peak (-58, -66, 2)

#4 2,552 mm³ t=-6.8
z=26 peak (38, -78, 26)



#5 2,394 mm³ t=-7.3
y=-82 peak (48, -82, 2)

#6 1,228 mm³ t=-6.2
x=-34 peak (-34, -36, -16)

#7 630 mm³ t=-7.1
z=-47 peak (18, -46, -47)

#8 598 mm³ t=-6.6
x=-34 peak (-34, 32, 40)

▲ Shannon surprise — Positive

2 connected components (18-conn) · 2 local maxima ≥8 mm apart · 13 voxels (0.0% of mask) · 0.4 cm³ · peak |t| = 6.0

#1 378 mm³ t=+6.0
y=-54 peak (-4, -54, 40)

#2 32 mm³ t=+5.2
x=-28 peak (-28, 24, 48)

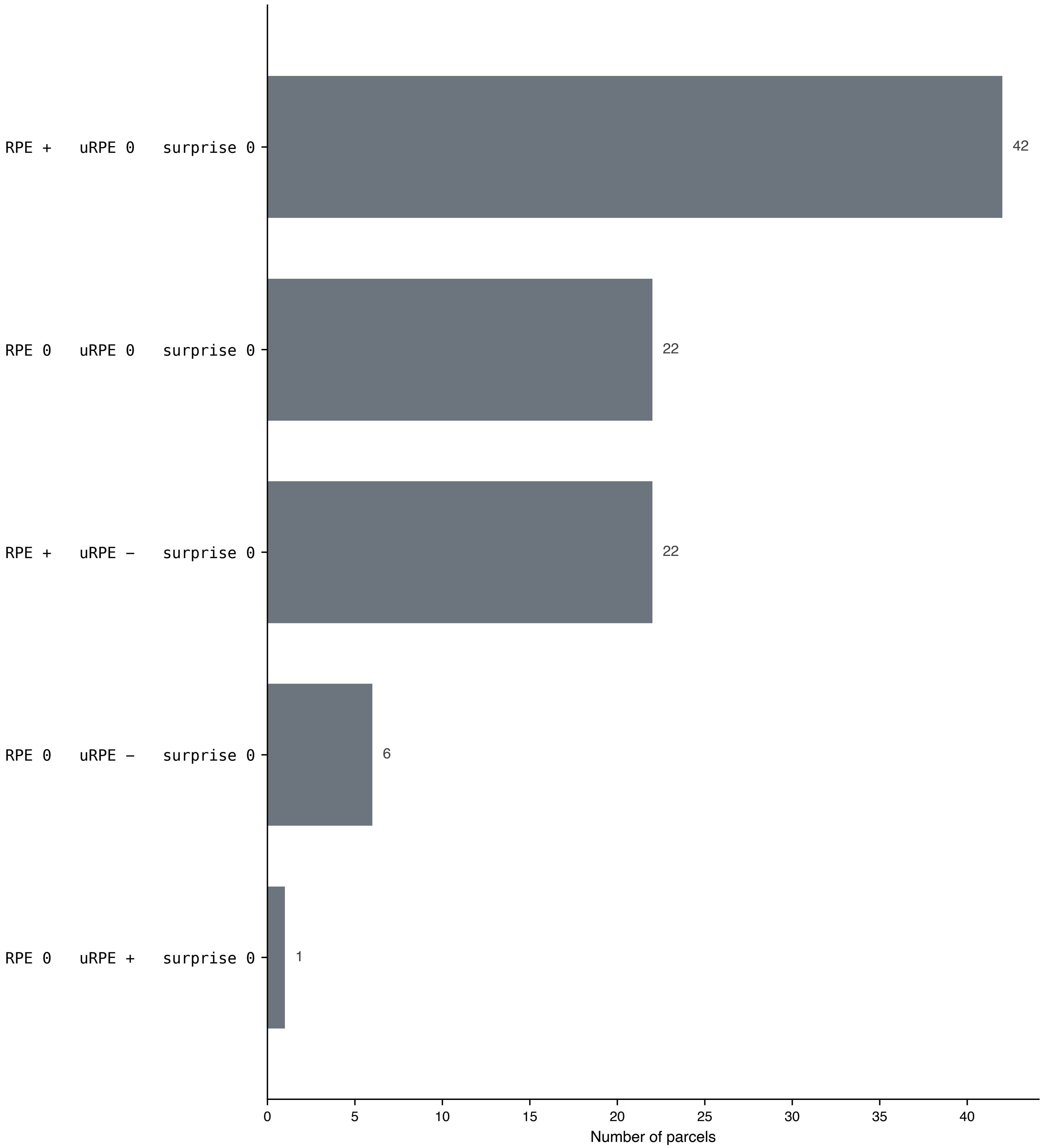
▼ Shannon surprise — Negative

0 connected components (18-conn) · 0 local maxima ≥8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm³ · peak |t| = 0.0

No suprathreshold clusters

Sign patterns across all relevant parcels — model7

93 Harvard-Oxford parcels are touched by at least one signal's surviving clusters at cluster-forming t > 3.24. For each parcel the sign of each signal is + / − / 0 by a Holm-corrected one-sample t-test on the parcel mean. Order below is RPE, uRPE, surprise.

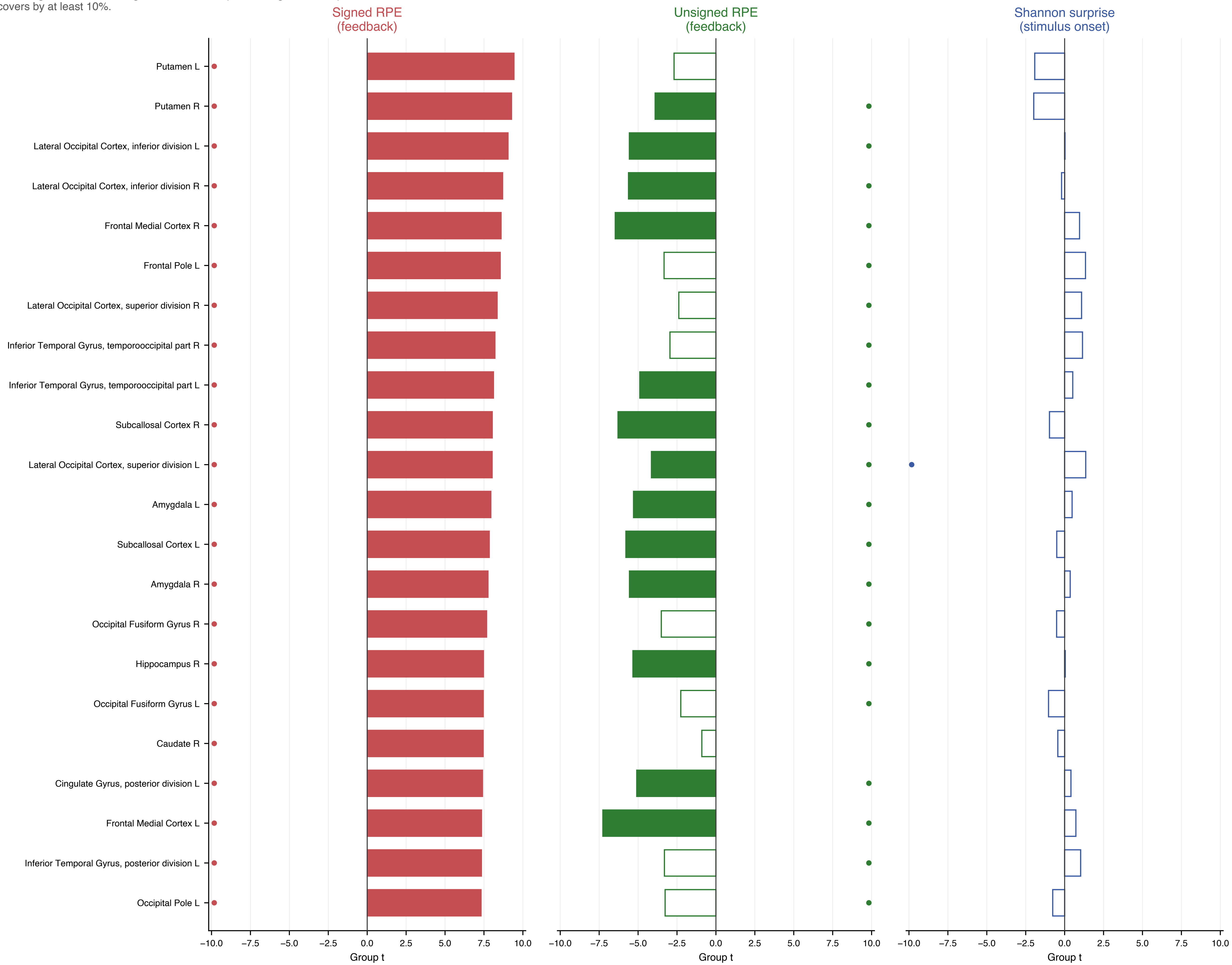


The single most common pattern is + 0 0 (42 of 93 parcels).
RPE is significantly positive in 64 parcels; uRPE is significantly negative in 28; the two co-occur in 22.
Surprise reaches significance in either direction in only 0 parcels.

Read this together with the orthogonalisation caveat: in model7 uRPE is the FIRST feedback modulator and signed RPE the second, so SPM's default serial orthogonalisation gives uRPE every scrap of variance the two share. A negative uRPE loading therefore cannot be an artefact of the ordering -- the ordering biases uRPE the other way.

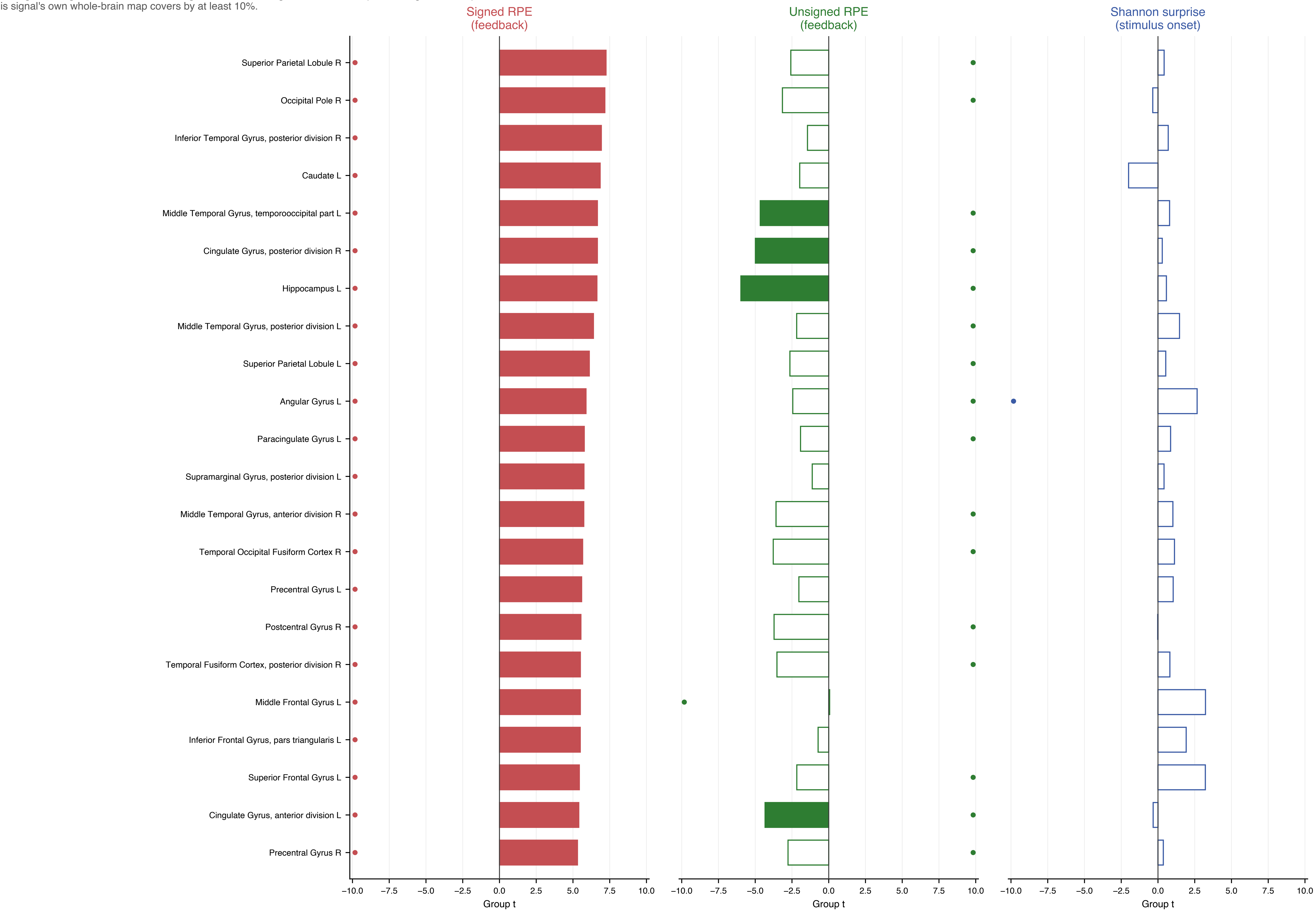
Which regions code which signal, and with which sign — model7 (1/5)

Harvard-Oxford parcels touched by any signal's surviving clusters (cluster-forming t > 3.24, p < .001). Bars are the group one-sample t of that parcel's mean contrast value; anatomical parcels give no signal a home-field advantage. Filled = Holm-significant across all parcel × signal tests, open = not. A dot on the far side of the axis marks parcels that this signal's own whole-brain map covers by at least 10%.



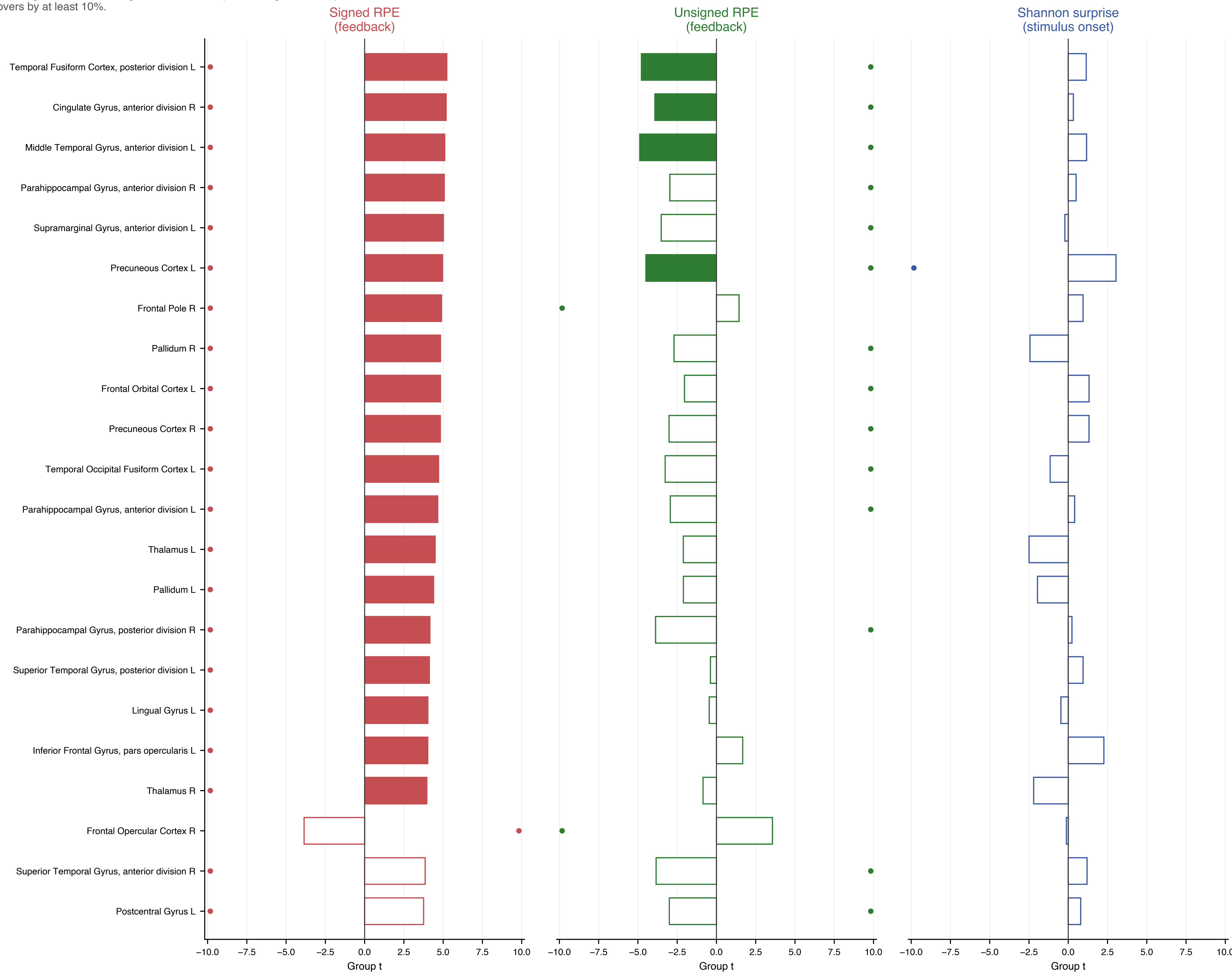
Which regions code which signal, and with which sign — model7 (2/5)

Harvard-Oxford parcels touched by any signal's surviving clusters (cluster-forming t > 3.24, p < .001). Bars are the group one-sample t of that parcel's mean contrast value; anatomical parcels give no signal a home-field advantage. Filled = Holm-significant across all parcel × signal tests, open = not. A dot on the far side of the axis marks parcels that this signal's own whole-brain map covers by at least 10%.



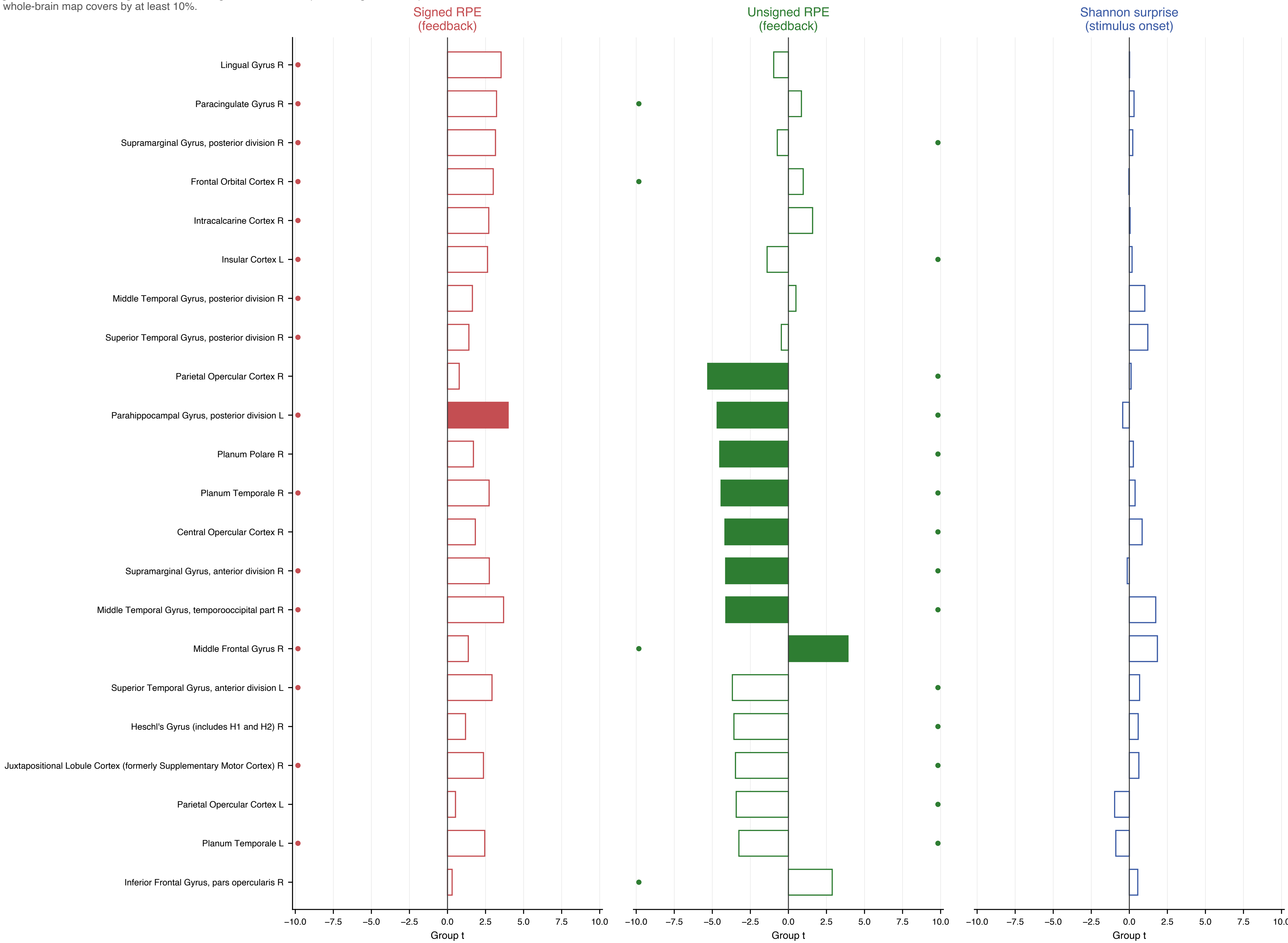
Which regions code which signal, and with which sign — model7 (3/5)

Harvard-Oxford parcels touched by any signal's surviving clusters (cluster-forming t > 3.24, p < .001). Bars are the group one-sample t of that parcel's mean contrast value; anatomical parcels give no signal a home-field advantage. Filled = Holm-significant across all parcel × signal tests, open = not. A dot on the far side of the axis marks parcels that this signal's own whole-brain map covers by at least 10%.



Which regions code which signal, and with which sign — model7 (4/5)

Harvard-Oxford parcels touched by any signal's surviving clusters (cluster-forming t > 3.24, p < .001). Bars are the group one-sample t of that parcel's mean contrast value; anatomical parcels give no signal a home-field advantage. Filled = Holm-significant across all parcel × signal tests, open = not. A dot on the far side of the axis marks parcels that this signal's own whole-brain map covers by at least 10%.



Which regions code which signal, and with which sign — model7 (5/5)

Harvard-Oxford parcels touched by any signal's surviving clusters (cluster-forming t > 3.24, p < .001). Bars are the group one-sample t of that parcel's mean contrast value; anatomical parcels give no signal a home-field advantage. Filled = Holm-significant across all parcel × signal tests, open = not. A dot on the far side of the axis marks parcels that this signal's own whole-brain map covers by at least 10%.